Introduction

Welcome to The Embedded Rust Book: An introductory book about using the Rust Programming Language on "Bare Metal" embedded systems, such as Microcontrollers.

Who Embedded Rust is For

Embedded Rust is for everyone who wants to do embedded programming while taking advantage of the higher-level concepts and safety guarantees the Rust language provides. (See also Who Rust Is For)

Scope

The goals of this book are:

- Get developers up to speed with embedded Rust development. i.e. How to set up a development environment.
- Share *current* best practices about using Rust for embedded development. i.e. How to best use Rust language features to write more correct embedded software.
- Serve as a cookbook in some cases. e.g. How do I mix C and Rust in a single project?

This book tries to be as general as possible but to make things easier for both the readers and the writers it uses the ARM Cortex-M architecture in all its examples. However, the book doesn't assume that the reader is familiar with this particular architecture and explains details particular to this architecture where required.

Who This Book is For

This book caters towards people with either some embedded background or some Rust background, however we believe everybody curious about embedded Rust programming can get something out of this book. For those without any prior knowledge we suggest you read the "Assumptions and Prerequisites" section and catch up on missing knowledge to get more out of the book and improve your reading experience. You can check out the "Other Resources" section to find resources on topics you might want to catch up on.

Assumptions and Prerequisites

- You are comfortable using the Rust Programming Language, and have written, run, and debugged Rust applications on a desktop environment. You should also be familiar with the idioms of the 2018 edition as this book targets Rust 2018.
- You are comfortable developing and debugging embedded systems in another language such as C, C++, or Ada, and are familiar with concepts such as:
  - Cross Compilation
  - Memory Mapped Peripherals
  - Interrupts
  - Common interfaces such as I2C, SPI, Serial, etc.

Other Resources

If you are unfamiliar with anything mentioned above or if you want more information about a specific topic mentioned in this book you might find some of these resources helpful.

Topic	Resource	Description
Rust	Rust Book	If you are not yet comfortable with Rust, we highly suggest reading this book.
Rust, Embedded	Discovery Book	If you have never done any embedded programming, this book might be a better start
Rust, Embedded	Embedded Rust Bookshelf	Here you can find several other resources provided by Rust's Embedded Working Group.
Rust, Embedded	Embedonomicon	The nitty gritty details when doing embedded programming in Rust.
Rust, Embedded	embedded FAQ	Frequently asked questions about Rust in an embedded context.
Rust, Embedded	Comprehensive Rust 🦀: Bare Metal	Teaching material for a 1-day class on bare-metal Rust development
Interrupts	Interrupt	-
Memory-mapped IO/Peripherals	Memory-mapped I/O	-

Topic	Resource	Description
SPI, UART, RS232, USB, I2C, TTL	Stack Exchange about SPI, UART, and other interfaces	-

Translations

This book has been translated by generous volunteers. If you would like your translation listed here, please open a PR to add it.

- Japanese (repository)
- Chinese (repository)

How to Use This Book

This book generally assumes that you're reading it front-to-back. Later chapters build on concepts in earlier chapters, and earlier chapters may not dig into details on a topic, revisiting the topic in a later chapter.

This book will be using the STM32F3DISCOVERY development board from STMicroelectronics for the majority of the examples contained within. This board is based on the ARM Cortex-M architecture, and while basic functionality is the same across most CPUs based on this architecture, peripherals and other implementation details of Microcontrollers are different between different vendors, and often even different between Microcontroller families from the same vendor.

For this reason, we suggest purchasing the STM32F3DISCOVERY development board for the purpose of following the examples in this book.

Contributing to This Book

The work on this book is coordinated in this repository and is mainly developed by the resources team.

If you have trouble following the instructions in this book or find that some section of the book is not clear enough or hard to follow then that's a bug and it should be reported in the issue tracker of this book.

Pull requests fixing typos and adding new content are very welcome!

Re-using this material

This book is distributed under the following licenses:

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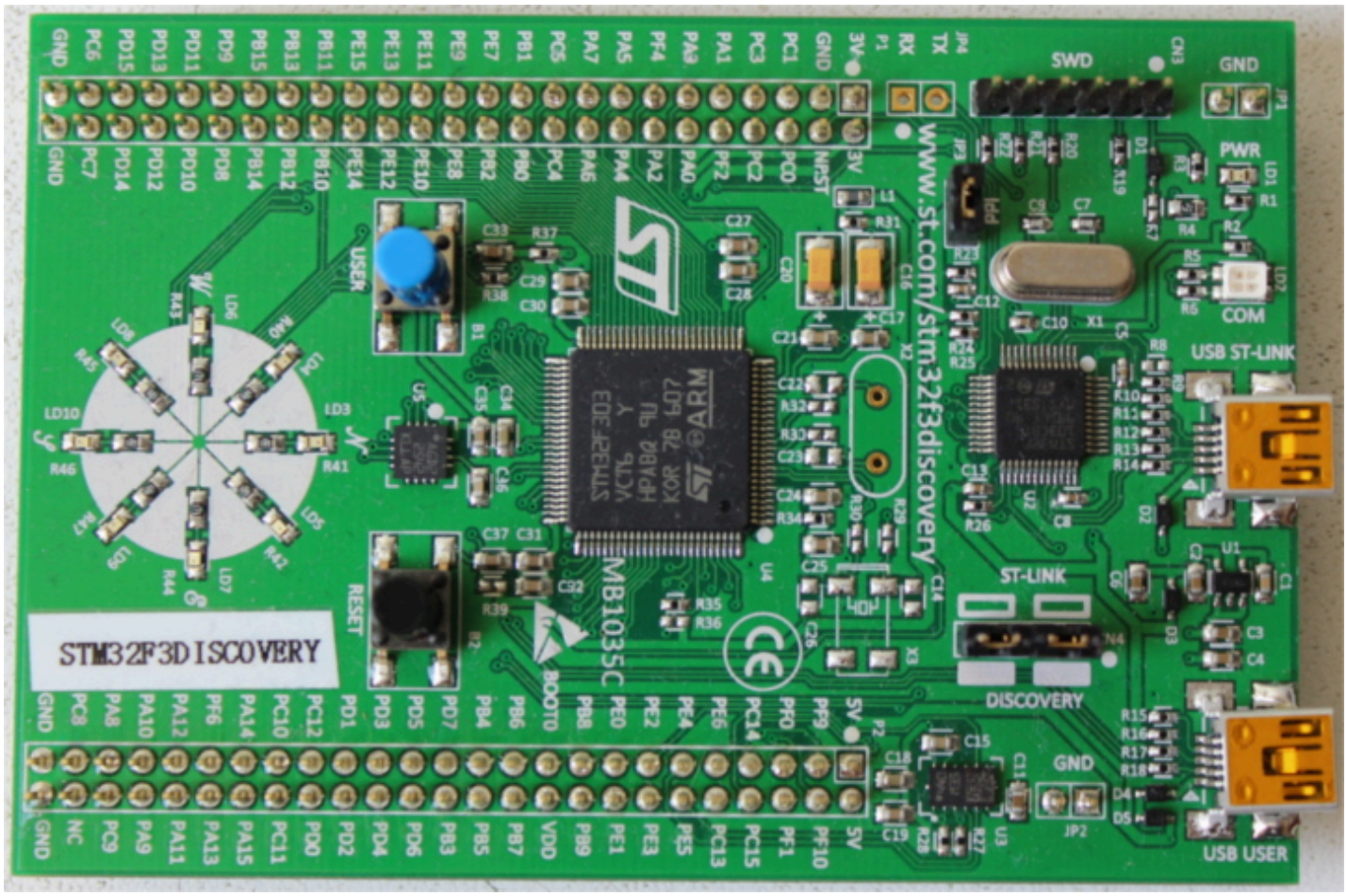
Meet Your Hardware

Let's get familiar with the hardware we'll be working with.

STM32F3DISCOVERY (the "F3")



What does this board contain?

- A STM32F303VCT6 microcontroller. This microcontroller has
  - A single-core ARM Cortex-M4F processor with hardware support for single-precision floating point operations and a maximum clock frequency of 72 MHz.
  - 256 KiB of "Flash" memory. (1 KiB = 10**24** bytes)
  - 48 KiB of RAM.
  - A variety of integrated peripherals such as timers, I2C, SPI and USART.

    - General purpose Input Output (GPIO) and other types of pins accessible through the two rows of headers along side the board.
    - A USB interface accessible through the USB port labeled "USB USER".
- An accelerometer as part of the LSM303DLHC chip.
- A magnetometer as part of the LSM303DLHC chip.
- A gyroscope as part of the L3GD20 chip.
- 8 user LEDs arranged in the shape of a compass.
- A second microcontroller: a STM32F103. This microcontroller is actually part of an on-board programmer / debugger and is connected to the USB port named "USB ST-LINK".

For a more detailed list of features and further specifications of the board take a look at the STMicroelectronics website.

A word of caution: be careful if you want to apply external signals to the board. The microcontroller STM32F303VCT6 pins take a nominal voltage of 3.3 volts. For further information consult the 6.2 Absolute maximum ratings section in the manual

A `no_std` Rust Environment

The term Embedded Programming is used for a wide range of different classes of programming. Ranging from programming 8-Bit MCUs (like the ST72325xx) with just a few KB of RAM and ROM, up to systems like the Raspberry Pi (Model B 3+) which has a 32/64-bit 4-core Cortex-A53 @ 1.4 GHz and 1GB of RAM. Different restrictions/limitations will apply when writing code depending on what kind of target and use case you have.

There are two general Embedded Programming classifications:

Hosted Environments

These kinds of environments are close to a normal PC environment. What this means is that you are provided with a System Interface E.G. POSIX that provides you with primitives to interact with various systems, such as file systems, networking, memory management, threads, etc. Standard libraries in turn usually depend on these primitives to implement their functionality. You may also have some sort of sysroot and restrictions on RAM/ROM-usage, and perhaps some special HW or I/Os. Overall it feels like coding on a special-purpose PC environment.

Bare Metal Environments

In a bare metal environment no code has been loaded before your program. Without the software provided by an OS we can not load the standard library. Instead the program, along with the crates it uses, can only use the hardware (bare metal) to run. To prevent rust from loading the standard library use `no_std`. The platform-agnostic parts of the standard library are available through libcore. libcore also excludes things which are not always desirable in an embedded environment. One of these things is a memory allocator for dynamic memory allocation. If you require this or any other functionalities there are often crates which provide these.

The libstd Runtime

As mentioned before using libstd requires some sort of system integration, but this is not only because libstd is just providing a common way of accessing OS abstractions, it also provides a runtime. This runtime, among other things, takes care of setting up stack overflow protection,

processing command line arguments, and spawning the main thread before a program's main function is invoked. This runtime also won't be available in a `no_std` environment.

Summary

`#![no_std]` is a crate-level attribute that indicates that the crate will link to the core-crate instead of the std-crate. The libcore crate in turn is a platform-agnostic subset of the std crate which makes no assumptions about the system the program will run on. As such, it provides APIs for language primitives like floats, strings and slices, as well as APIs that expose processor features like atomic operations and SIMD instructions. However it lacks APIs for anything that involves platform integration. Because of these properties no_std and libcore code can be used for any kind of bootstrapping (stage 0) code like bootloaders, firmware or kernels.

Overview

feature	no_std	std
heap (dynamic memory)	*	✓
collections (Vec, BTreeMap, etc)	**	✓
stack overflow protection	✗	✓
runs init code before main	✗	✓
libstd available	✗	✓
libcore available	✓	✓
writing firmware, kernel, or bootloader code	✓	✗

* Only if you use the `alloc` crate and use a suitable allocator like alloc-cortex-m.

** Only if you use the `collections` crate and configure a global default allocator.

** HashMap and HashSet are not available due to a lack of a secure random number generator.

See Also

- RFC-1184

Tooling

Dealing with microcontrollers involves using several different tools as we'll be dealing with an architecture different than your laptop's and we'll have to run and debug programs on a *remote* device.

We'll use all the tools listed below. Any recent version should work when a minimum version is not specified, but we have listed the versions we have tested.

- Rust 1.31, 1.31-beta, or a newer toolchain PLUS ARM Cortex-M compilation support.
- `cargo-binutils` ~0.1.4
- `qemu-system-arm` . Tested versions: 3.0.0
- OpenOCD >=0.8. Tested versions: v0.9.0 and v0.10.0
- GDB with ARM support. Version 7.12 or newer highly recommended. Tested versions: 7.10, 7.11, 7.12 and 8.1
- `cargo-generate` or `git` . These tools are optional but will make it easier to follow along with the book.

The text below explains why we are using these tools. Installation instructions can be found on the next page.

`cargo-generate` OR `git`

Bare metal programs are non-standard (`no_std`) Rust programs that require some adjustments to the linking process in order to get the memory layout of the program right. This requires some additional files (like linker scripts) and settings (like linker flags). We have packaged those for you in a template such that you only need to fill in the missing information (such as the project name and the characteristics of your target hardware).

Our template is compatible with `cargo-generate` : a Cargo subcommand for creating new Cargo projects from templates. You can also download the template using `git` , `curl` , `wget` , or your web browser.

`cargo-binutils`

`cargo-binutils` is a collection of Cargo subcommands that make it easy to use the LLVM tools that are shipped with the Rust toolchain. These tools include the LLVM versions of `objdump` ,

`nm` and `size` and are used for inspecting binaries.

The advantage of using these tools over GNU binutils is that (a) installing the LLVM tools is the same one-command installation (`rustup component add llvm-tools`) regardless of your OS and (b) tools like `objdump` support all the architectures that `rustc` supports -- from ARM to x86_64 -- because they both share the same LLVM backend.

`qemu-system-arm`

QEMU is an emulator. In this case we use the variant that can fully emulate ARM systems. We use QEMU to run embedded programs on the host. Thanks to this you can follow some parts of this book even if you don't have any hardware with you!

Tooling for Embedded Rust Debugging

Overview

Debugging embedded systems in Rust requires specialized tools including software to manage the debugging process, debuggers to inspect and control program execution, and hardware probes to facilitate interaction between the host and the embedded device. This document outlines essential software tools like Probe-rs and OpenOCD, which simplify and support the debugging process, alongside prominent debuggers such as GDB and the Probe-rs Visual Studio Code extension. Additionally, it covers key hardware probes such as Rusty-probe, ST-Link, J-Link, and MCU-Link, which are integral for effective debugging and programming of embedded devices.

Software that drives debugging tools

Probe-rs

Probe-rs is a modern, Rust-focused software designed to work with debuggers in embedded systems. Unlike OpenOCD, Probe-rs is built with simplicity in mind and aims to reduce the configuration burden often found in other debugging solutions. It supports various probes and targets, providing a high-level interface for interacting with embedded hardware. Probe-rs

integrates directly with Rust tooling, and integrates with Visual Studio Code through its extension, allowing developers to streamline their debugging workflow.

OpenOCD (Open On-Chip Debugger)

OpenOCD is an open-source software tool used for debugging, testing, and programming embedded systems. It provides an interface between the host system and embedded hardware, supporting various transport layers like JTAG and SWD (Serial Wire Debug). OpenOCD integrates with GDB, which is a debugger. OpenOCD is widely supported, with extensive documentation and a large community, but may require complex configuration, especially for custom embedded setups.

Debuggers

A debugger allows developers to inspect and control the execution of a program in order to identify and correct errors or bugs. It provides functionalities such as setting breakpoints, stepping through code line by line, and examining the values of variables and memory states. Debuggers are essential for thorough software development and maintenance, enabling developers to ensure that their code behaves as intended under various conditions.

Debuggers know how to:

- Interact with the memory mapped registers.
- Set Breakpoints/Watchpoints.
- Read and write to the memory mapped registers.
- Detect when the MCU has been halted for a debug event.
- Continue MCU execution after a debug event has been encountered.
- Erase and write to the microcontroller's FLASH.

Probe-rs Visual Studio Code Extension

Probe-rs has a Visual Studio Code extension, providing a seamless debugging experience without extensive setup. Through this connection, developers can use Rust-specific features like pretty printing and detailed error messages, ensuring that their debugging process aligns with the Rust ecosystem.

GDB (GNU Debugger)

GDB is a versatile debugging tool that allows developers to examine the state of programs while they run or after they crash. For embedded Rust, GDB connects to the target system via OpenOCD or other debugging servers to interact with the embedded code. GDB is highly configurable and supports features like remote debugging, variable inspection, and conditional breakpoints. It can be used on a variety of platforms, and has extensive support for Rust-specific debugging needs, such as pretty printing and integration with IDEs.

Probes

A hardware probe is a device used in the development and debugging of embedded systems to facilitate communication between a host computer and the target embedded device. It typically supports protocols like JTAG or SWD, enabling it to program, debug, and analyze the microcontroller or microprocessor on the embedded system. Hardware probes are crucial for developers to set breakpoints, step through code, and inspect memory and processor registers, effectively allowing them to diagnose and fix issues in real-time.

Rusty-probe

Rusty-probe is an open-sourced USB-based hardware debugging probe designed to work with probe-rs. The combination of Rusty-Probe and probe-rs provides an easy-to-use, cost-effective solution for developers working with embedded Rust applications.

ST-Link

The ST-Link is a popular debugging and programming probe developed by STMicroelectronics primarily for their STM32 and STM8 microcontroller series. It supports both debugging and programming via JTAG or SWD (Serial Wire Debug) interfaces. ST-Link is widely used due to its direct support from STMicroelectronics' extensive range of development boards and its integration into major IDEs, making it a convenient choice for developers working with STM microcontrollers.

J-Link

J-Link, developed by SEGGER Microcontroller, is a robust and versatile debugger supporting a wide range of CPU cores and devices beyond just ARM, such as RISC-V. Known for its high

performance and reliability, J-Link supports various communication interfaces, including JTAG, SWD, and fine-pitch JTAG interfaces. It is favored for its advanced features like unlimited breakpoints in flash memory and its compatibility with a multitude of development environments.

MCU-Link

MCU-Link is a debugging probe that also functions as a programmer, provided by NXP Semiconductors. It supports a variety of ARM Cortex microcontrollers and interfaces seamlessly with development tools like MCUXpresso IDE. MCU-Link is particularly notable for its versatility and affordability, making it an accessible option for hobbyists, educators, and professional developers alike.

Installing the tools

This page contains OS-agnostic installation instructions for a few of the tools:

Rust Toolchain

Install rustup by following the instructions at https://rustup.rs.

NOTE Make sure you have a compiler version equal to or newer than `1.31`. `rustc -V` should return a date newer than the one shown below.

```
$ rustc -V
rustc 1.31.1 (b6c32da9b 2018-12-18)
```

For bandwidth and disk usage concerns the default installation only supports native compilation. To add cross compilation support for the ARM Cortex-M architectures choose one of the following compilation targets. For the STM32F3DISCOVERY board used for the examples in this book, use the `thumbv7em-none-eabihf` target. Find the best Cortex-M for you.

Cortex-M0, M0+, and M1 (ARMv6-M architecture):

```
rustup target add thumbv6m-none-eabi
```

Cortex-M3 (ARMv7-M architecture):

```
rustup target add thumbv7m-none-eabi
```

Cortex-M4 and M7 without hardware floating point (ARMv7E-M architecture):

```
rustup target add thumbv7em-none-eabi
```

Cortex-M4F and M7F with hardware floating point (ARMv7E-M architecture):

```
rustup target add thumbv7em-none-eabihf
```

Cortex-M23 (ARMv8-M architecture):

```
rustup target add thumbv8m.base-none-eabi
```

Cortex-M33 and M35P (ARMv8-M architecture):

```
rustup target add thumbv8m.main-none-eabi
```

Cortex-M33F and M35PF with hardware floating point (ARMv8-M architecture):

```
rustup target add thumbv8m.main-none-eabihf
```

`cargo-binutils`

```
cargo install cargo-binutils
```

```
rustup component add llvm-tools
```

WINDOWS: prerequisite C++ Build Tools for Visual Studio 2019 is installed. https://visualstudio.microsoft.com/thank-you-downloading-visual-studio/?sku=BuildTools&rel=16

`cargo-generate`

We'll use this later to generate a project from a template.

```
cargo install cargo-generate
```

Note: on some Linux distros (e.g. Ubuntu) you may need to install the packages `libssl-dev` and `pkg-config` prior to installing cargo-generate.

OS-Specific Instructions

Now follow the instructions specific to the OS you are using:

- Linux
- Windows
- macOS

Linux

Here are the installation commands for a few Linux distributions.

Packages

- Ubuntu 18.04 or newer / Debian stretch or newer

NOTE `gdb-multiarch` is the GDB command you'll use to debug your ARM Cortex-M programs

---

```
sudo apt install gdb-multiarch openocd qemu-system-arm
```

- Ubuntu 14.04 and 16.04

NOTE `arm-none-eabi-gdb` is the GDB command you'll use to debug your ARM Cortex-M programs

---

```
sudo apt install gdb-arm-none-eabi openocd qemu-system-arm
```

- Fedora 27 or newer

```
sudo dnf install gdb openocd qemu-system-arm
```

- Arch Linux

NOTE `arm-none-eabi-gdb` is the GDB command you'll use to debug ARM Cortex-M programs

---

```
sudo pacman -S arm-none-eabi-gdb qemu-system-arm openocd
```

udev rules

This rule lets you use OpenOCD with the Discovery board without root privilege.

Create the file `/etc/udev/rules.d/70-st-link.rules` with the contents shown below.

```
# STM32F3DISCOVERY rev A/B - ST-LINK/V2
ATTRS{idVendor}=="0483", ATTRS{idProduct}=="3748", TAG+="uaccess"

# STM32F3DISCOVERY rev C+ - ST-LINK/V2-1
ATTRS{idVendor}=="0483", ATTRS{idProduct}=="374b", TAG+="uaccess"
```

Then reload all the udev rules with:

```
sudo udevadm control --reload-rules
```

If you had the board plugged to your laptop, unplug it and then plug it again.

You can check the permissions by running this command:

```
lsusb
```

Which should show something like

```
(..)
Bus 001 Device 018: ID 0483:374b STMicroelectronics ST-LINK/V2.1
(..)
```

Take note of the bus and device numbers. Use those numbers to create a path like `/dev/bus/usb/<bus>/<device>`. Then use this path like so:

```
ls -l /dev/bus/usb/001/018
```

```
crw-------+ 1 root root 189, 17 Sep 13 12:34 /dev/bus/usb/001/018
```

```
getfacl /dev/bus/usb/001/018 | grep user
```

```
user::rw-
user:you:rw-
```

The `+` appended to permissions indicates the existence of an extended permission. The `getfacl` command tells the user `you` can make use of this device.

Now, go to the next section.

macOS

All the tools can be installed using Homebrew or MacPorts:

Install tools with Homebrew

```
$ # GDB
$ brew install arm-none-eabi-gdb

$ # OpenOCD
$ brew install openocd

$ # QEMU
$ brew install qemu
```

NOTE If OpenOCD crashes you may need to install the latest version using:

```
$ brew install --HEAD openocd
```

Install tools with MacPorts

```
$ # GDB
$ sudo port install arm-none-eabi-gcc

$ # OpenOCD
$ sudo port install openocd

$ # QEMU
$ sudo port install qemu
```

That's all! Go to the next section.

Windows

arm-none-eabi-gdb

ARM provides `.exe` installers for Windows. Grab one from here, and follow the instructions. Just before the installation process finishes tick/select the "Add path to environment variable" option. Then verify that the tools are in your `%PATH%`:

```
$ arm-none-eabi-gdb -v
GNU gdb (GNU Tools for Arm Embedded Processors 7-2018-q2-update) 8.1.0.20180315-
git
(..)
```

OpenOCD

There's no official binary release of OpenOCD for Windows but if you're not in the mood to compile it yourself, the xPack project provides a binary distribution, here. Follow the provided installation instructions. Then update your `%PATH%` environment variable to include the path where the binaries were installed. (`C:\Users\USERNAME\AppData\Roaming\xPacks\@xpack-dev-tools\openocd\0.10.0-13.1\.content\bin\` , if you've been using the easy install)

Verify that OpenOCD is in your `%PATH%` with:

```
$ openocd -v
Open On-Chip Debugger 0.10.0
(..)
```

QEMU

Grab QEMU from the official website.

ST-LINK USB driver

You'll also need to install this USB driver or OpenOCD won't work. Follow the installer instructions and make sure you install the right version (32-bit or 64-bit) of the driver.

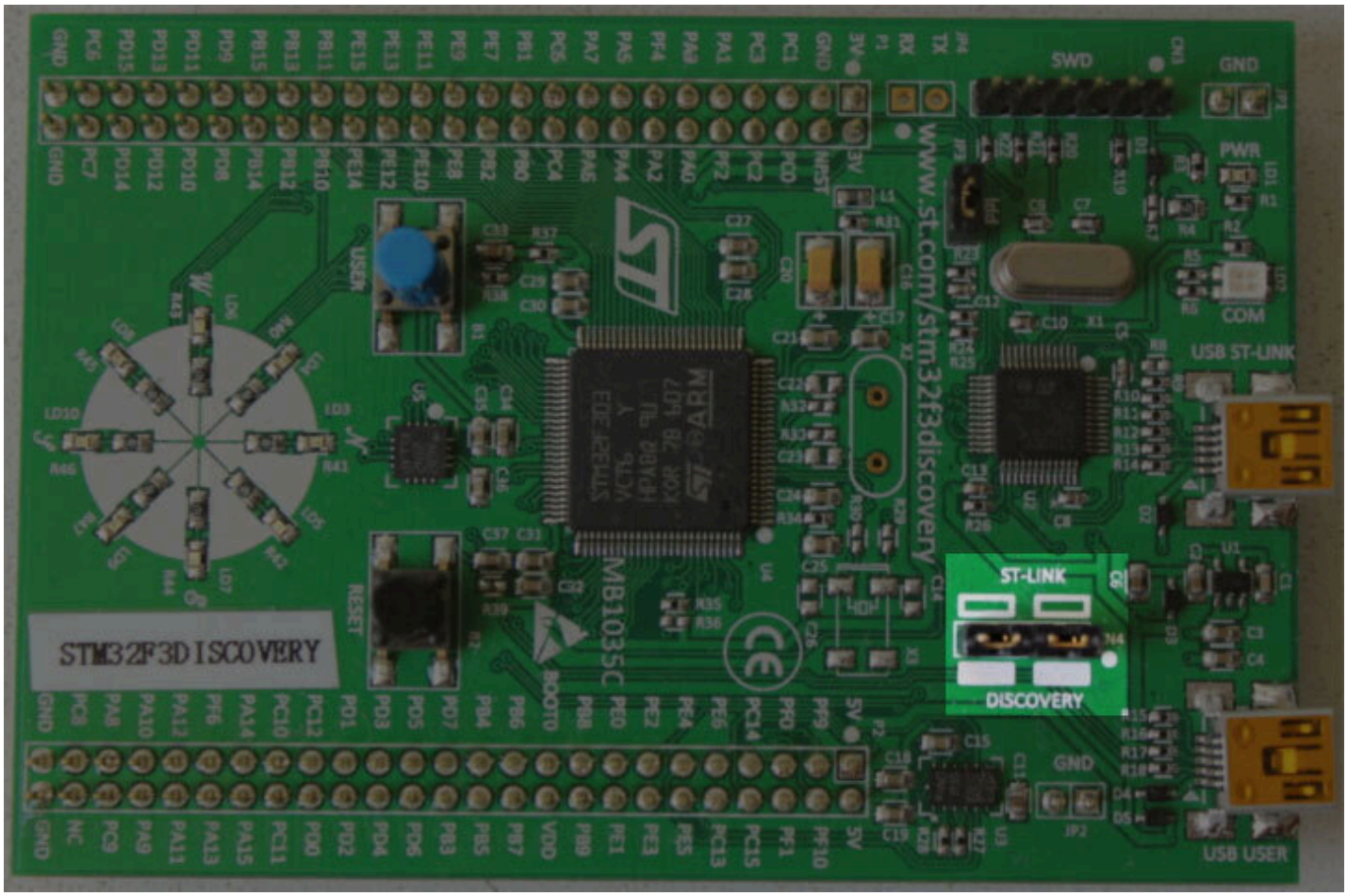
That's all! Go to the next section.

Verify Installation

In this section we check that some of the required tools / drivers have been correctly installed and configured.

Connect your laptop / PC to the discovery board using a Mini-USB USB cable. The discovery board has two USB connectors; use the one labeled "USB ST-LINK" that sits on the center of the edge of the board.

Also check that the ST-LINK header is populated. See the picture below; the ST-LINK header is highlighted.



Now run the following command:

```
openocd -f interface/stlink.cfg -f target/stm32f3x.cfg
```

NOTE: Old versions of openocd, including the 0.10.0 release from 2017, do not contain the new (and preferable) `interface/stlink.cfg` file; instead you may need to use

`interface/stlink-v2.cfg` or `interface/stlink-v2-1.cfg`.

You should get the following output and the program should block the console:

```
Open On-Chip Debugger 0.10.0
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
Info : auto-selecting first available session transport "hla_swd". To override use
'transport select <transport>'.
adapter speed: 1000 kHz
adapter_nsrst_delay: 100
Info : The selected transport took over low-level target control. The results
might differ compared to plain JTAG/SWD
none separate
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : clock speed 950 kHz
Info : STLINK v2 JTAG v27 API v2 SWIM v15 VID 0x0483 PID 0x374B
Info : using stlink api v2
Info : Target voltage: 2.919881
Info : stm32f3x.cpu: hardware has 6 breakpoints, 4 watchpoints
```

The contents may not match exactly but you should get the last line about breakpoints and watchpoints. If you got it then terminate the OpenOCD process and move to the next section.

If you didn't get the "breakpoints" line then try one of the following commands.

```
openocd -f interface/stlink-v2.cfg -f target/stm32f3x.cfg
```

```
openocd -f interface/stlink-v2-1.cfg -f target/stm32f3x.cfg
```

If one of those commands works it means you got an old hardware revision of the discovery board. That won't be a problem but commit that fact to memory as you'll need to configure things a bit differently later on. You can move to the next section.

If none of the commands work as a normal user then try to run them with root permission (e.g. `sudo openocd ..`). If the commands do work with root permission then check that the udev rules have been correctly set.

If you have reached this point and OpenOCD is not working please open an issue and we'll help you out!

Getting Started

In this section we'll walk you through the process of writing, building, flashing and debugging embedded programs. You will be able to try most of the examples without any special hardware as we will show you the basics using QEMU, a popular open-source hardware emulator. The only section where hardware is required is, naturally enough, the Hardware section, where we use OpenOCD to program an STM32F3DISCOVERY.

QEMU

We'll start writing a program for the LM3S6965, a Cortex-M3 microcontroller. We have chosen this as our initial target because it can be emulated using QEMU so you don't need to fiddle with hardware in this section and we can focus on the tooling and the development process.

IMPORTANT We'll use the name "app" for the project name in this tutorial. Whenever you see the word "app" you should replace it with the name you selected for your project. Or, you could also name your project "app" and avoid the substitutions.

Creating a non standard Rust program

We'll use the `cortex-m-quickstart` project template to generate a new project from it. The created project will contain a barebone application: a good starting point for a new embedded rust application. In addition, the project will contain an `examples` directory, with several separate applications, highlighting some of the key embedded rust functionality.

Using `cargo-generate`

First install cargo-generate

```
cargo install cargo-generate
```

Then generate a new project

```
cargo generate --git https://github.com/rust-embedded/cortex-m-quickstart
```

```
Project Name: app
Creating project called `app`...
Done! New project created /tmp/app
```

```
cd app
```

Using `git`

Clone the repository

```
git clone https://github.com/rust-embedded/cortex-m-quickstart app
cd app
```

And then fill in the placeholders in the `Cargo.toml` file

```
[package]
authors = ["{{authors}}"] # "{{authors}}" -> "John Smith"
edition = "2018"
name = "{{project-name}}" # "{{project-name}}" -> "app"
version = "0.1.0"

# ..

[[bin]]
name = "{{project-name}}" # "{{project-name}}" -> "app"
test = false
bench = false
```

Using neither

Grab the latest snapshot of the `cortex-m-quickstart` template and extract it.

```
curl -LO https://github.com/rust-embedded/cortex-m-quickstart/archive/master.zip
unzip master.zip
mv cortex-m-quickstart-master app
cd app
```

Or you can browse to `cortex-m-quickstart`, click the green "Clone or download" button and then click "Download ZIP".

Then fill in the placeholders in the `Cargo.toml` file as done in the second part of the "Using `git`" version.

Program Overview

For convenience here are the most important parts of the source code in `src/main.rs`:

```
#![no_std]
#![no_main]

use panic_halt as _;

use cortex_m_rt::entry;

#[entry]
fn main() -> ! {
    loop {
        // your code goes here
    }
}
```

This program is a bit different from a standard Rust program so let's take a closer look.

`#![no_std]` indicates that this program will *not* link to the standard crate, `std`. Instead it will link to its subset: the `core` crate.

`#![no_main]` indicates that this program won't use the standard `main` interface that most Rust programs use. The main (no pun intended) reason to go with `no_main` is that using the `main` interface in `no_std` context requires nightly.

`use panic_halt as _;`. This crate provides a `panic_handler` that defines the panicking behavior of the program. We will cover this in more detail in the Panicking chapter of the book.

`#[entry]` is an attribute provided by the `cortex-m-rt` crate that's used to mark the entry point of the program. As we are not using the standard `main` interface we need another way to indicate the entry point of the program and that'd be `#[entry]`.

`fn main() -> !`. Our program will be the *only* process running on the target hardware so we don't want it to end! We use a divergent function (the `-> !` bit in the function signature) to ensure at compile time that'll be the case.

Cross compiling

The next step is to *cross* compile the program for the Cortex-M3 architecture. That's as simple as running `cargo build --target $TRIPLE` if you know what the compilation target (`$TRIPLE`) should be. Luckily, the `.cargo/config.toml` in the template has the answer:

```
tail -n6 .cargo/config.toml
```

```
[build]
# Pick ONE of these compilation targets
# target = "thumbv6m-none-eabi"    # Cortex-M0 and Cortex-M0+
target = "thumbv7m-none-eabi"    # Cortex-M3
# target = "thumbv7em-none-eabi"   # Cortex-M4 and Cortex-M7 (no FPU)
# target = "thumbv7em-none-eabihf" # Cortex-M4F and Cortex-M7F (with FPU)
```

To cross compile for the Cortex-M3 architecture we have to use `thumbv7m-none-eabi`. That target is not automatically installed when installing the Rust toolchain, it would now be a good time to add that target to the toolchain, if you haven't done it yet:

```
rustup target add thumbv7m-none-eabi
```

Since the `thumbv7m-none-eabi` compilation target has been set as the default in your `.cargo/config.toml` file, the two commands below do the same:

```
cargo build --target thumbv7m-none-eabi
cargo build
```

Inspecting

Now we have a non-native ELF binary in `target/thumbv7m-none-eabi/debug/app`. We can inspect it using `cargo-binutils`.

With `cargo-readobj` we can print the ELF headers to confirm that this is an ARM binary.

```
cargo readobj --bin app -- --file-headers
```

Note that:

- `--bin app` is sugar for inspect the binary at `target/$TRIPLE/debug/app`
- `--bin app` will also (re)compile the binary, if necessary

```
ELF Header:
  Magic:   7f 45 4c 46 01 01 01 00 00 00 00 00 00 00 00 00
  Class:                             ELF32
  Data:                              2's complement, little endian
  Version:                           1 (current)
  OS/ABI:                            UNIX - System V
  ABI Version:                       0x0
  Type:                              EXEC (Executable file)
  Machine:                           ARM
  Version:                           0x1
  Entry point address:               0x405
  Start of program headers:          52 (bytes into file)
  Start of section headers:          153204 (bytes into file)
  Flags:                             0x5000200
  Size of this header:               52 (bytes)
  Size of program headers:           32 (bytes)
  Number of program headers:         2
  Size of section headers:           40 (bytes)
  Number of section headers:         19
  Section header string table index: 18
```

`cargo-size` can print the size of the linker sections of the binary.

```
cargo size --bin app --release -- -A
```

we use `--release` to inspect the optimized version

```
app  :
section             size        addr
.vector_table       1024         0x0
.text                 92       0x400
.rodata                0       0x45c
.data                  0  0x20000000
.bss                   0  0x20000000
.debug_str          2958         0x0
.debug_loc            19         0x0
.debug_abbrev        567         0x0
.debug_info         4929         0x0
.debug_ranges         40         0x0
.debug_macinfo         1         0x0
.debug_pubnames     2035         0x0
.debug_pubtypes     1892         0x0
.ARM.attributes       46         0x0
.debug_frame         100         0x0
.debug_line          867         0x0
Total              14570
```

A refresher on ELF linker sections

- `.text` contains the program instructions
- `.rodata` contains constant values like strings
- `.data` contains statically allocated variables whose initial values are *not* zero
- `.bss` also contains statically allocated variables whose initial values *are* zero
- `.vector_table` is a *non*-standard section that we use to store the vector (interrupt) table
- `.ARM.attributes` and the `.debug_*` sections contain metadata and will *not* be loaded onto the target when flashing the binary.

IMPORTANT: ELF files contain metadata like debug information so their *size on disk* does *not* accurately reflect the space the program will occupy when flashed on a device. *Always* use `cargo-size` to check how big a binary really is.

`cargo-objdump` can be used to disassemble the binary.

```
cargo objdump --bin app --release -- --disassemble --no-show-raw-insn --print-imm-hex
```

> **NOTE** if the above command complains about `Unknown command line argument` see the following bug report: https://github.com/rust-embedded/book/issues/269

> **NOTE** this output can differ on your system. New versions of rustc, LLVM and libraries can generate different assembly. We truncated some of the instructions to keep the snippet small.

```
app:  file format ELF32-arm-little

Disassembly of section .text:
main:
     400: bl  #0x256
     404: b #-0x4 <main+0x4>

Reset:
     406: bl  #0x24e
     40a: movw  r0, #0x0
     < .. truncated any more instructions .. >

DefaultHandler_:
     656: b #-0x4 <DefaultHandler_>

UsageFault:
     657: strb  r7, [r4, #0x3]

DefaultPreInit:
     658: bx  lr

__pre_init:
     659: strb  r7, [r0, #0x1]

__nop:
     65a: bx  lr

HardFaultTrampoline:
     65c: mrs r0, msp
     660: b #-0x2 <HardFault_>

HardFault_:
     662: b #-0x4 <HardFault_>

HardFault:
     663: <unknown>
```

Running

Next, let's see how to run an embedded program on QEMU! This time we'll use the `hello` example which actually does something.

For convenience here's the source code of `examples/hello.rs`:

```rust
//! Prints "Hello, world!" on the host console using semihosting

#![no_main]
#![no_std]

use panic_halt as _;

use cortex_m_rt::entry;
use cortex_m_semihosting::{debug, hprintln};

#[entry]
fn main() -> ! {
    hprintln!("Hello, world!").unwrap();

    // exit QEMU
    // NOTE do not run this on hardware; it can corrupt OpenOCD state
    debug::exit(debug::EXIT_SUCCESS);

    loop {}
}
```

This program uses something called semihosting to print text to the *host* console. When using real hardware this requires a debug session but when using QEMU this Just Works.

Let's start by compiling the example:

```
cargo build --example hello
```

The output binary will be located at `target/thumbv7m-none-eabi/debug/examples/hello`.

To run this binary on QEMU run the following command:

```
qemu-system-arm \
  -cpu cortex-m3 \
  -machine lm3s6965evb \
  -nographic \
  -semihosting-config enable=on,target=native \
  -kernel target/thumbv7m-none-eabi/debug/examples/hello
```

```
Hello, world!
```

The command should successfully exit (exit code = 0) after printing the text. On *nix you can check that with the following command:

```
echo $?
```

```
0
```

Let's break down that QEMU command:

- `qemu-system-arm` . This is the QEMU emulator. There are a few variants of these QEMU binaries; this one does full *system* emulation of *ARM* machines hence the name.
- `-cpu cortex-m3` . This tells QEMU to emulate a Cortex-M3 CPU. Specifying the CPU model lets us catch some miscompilation errors: for example, running a program compiled for the Cortex-M4F, which has a hardware FPU, will make QEMU error during its execution.
- `-machine lm3s6965evb` . This tells QEMU to emulate the LM3S6965EVB, an evaluation board that contains a LM3S6965 microcontroller.
- `-nographic` . This tells QEMU to not launch its GUI.
- `-semihosting-config (..)` . This tells QEMU to enable semihosting. Semihosting lets the emulated device, among other things, use the host stdout, stderr and stdin and create files on the host.
- `-kernel $file` . This tells QEMU which binary to load and run on the emulated machine.

Typing out that long QEMU command is too much work! We can set a custom runner to simplify the process. `.cargo/config.toml` has a commented out runner that invokes QEMU; let's uncomment it:

```
head -n3 .cargo/config.toml
```

```
[target.thumbv7m-none-eabi]
# uncomment this to make `cargo run` execute programs on QEMU
runner = "qemu-system-arm -cpu cortex-m3 -machine lm3s6965evb -nographic -
semihosting-config enable=on,target=native -kernel"
```

This runner only applies to the `thumbv7m-none-eabi` target, which is our default compilation target. Now `cargo run` will compile the program and run it on QEMU:

```
cargo run --example hello --release
```

```
   Compiling app v0.1.0 (file:///tmp/app)
    Finished release [optimized + debuginfo] target(s) in 0.26s
     Running `qemu-system-arm -cpu cortex-m3 -machine lm3s6965evb -nographic -
semihosting-config enable=on,target=native -kernel target/thumbv7m-none-
eabi/release/examples/hello`
Hello, world!
```

Debugging

Debugging is critical to embedded development. Let's see how it's done.

Debugging an embedded device involves *remote* debugging as the program that we want to debug won't be running on the machine that's running the debugger program (GDB or LLDB).

Remote debugging involves a client and a server. In a QEMU setup, the client will be a GDB (or LLDB) process and the server will be the QEMU process that's also running the embedded program.

In this section we'll use the `hello` example we already compiled.

The first debugging step is to launch QEMU in debugging mode:

```
qemu-system-arm \
  -cpu cortex-m3 \
  -machine lm3s6965evb \
  -nographic \
  -semihosting-config enable=on,target=native \
  -gdb tcp::3333 \
  -S \
  -kernel target/thumbv7m-none-eabi/debug/examples/hello
```

This command won't print anything to the console and will block the terminal. We have passed two extra flags this time:

- `-gdb tcp::3333` . This tells QEMU to wait for a GDB connection on TCP port 3333.
- `-S` . This tells QEMU to freeze the machine at startup. Without this the program would have reached the end of main before we had a chance to launch the debugger!

Next we launch GDB in another terminal and tell it to load the debug symbols of the example:

```
gdb-multiarch -q target/thumbv7m-none-eabi/debug/examples/hello
```

NOTE: you might need another version of gdb instead of `gdb-multiarch` depending on which one you installed in the installation chapter. This could also be `arm-none-eabi-gdb` or just `gdb`.

Then within the GDB shell we connect to QEMU, which is waiting for a connection on TCP port 3333.

```
target remote :3333
```

```
Remote debugging using :3333
Reset () at $REGISTRY/cortex-m-rt-0.6.1/src/lib.rs:473
473     pub unsafe extern "C" fn Reset() -> ! {
```

You'll see that the process is halted and that the program counter is pointing to a function named `Reset`. That is the reset handler: what Cortex-M cores execute upon booting.

Note that on some setup, instead of displaying the line `Reset () at $REGISTRY/cortex-m-rt-0.6.1/src/lib.rs:473` as shown above, gdb may print some warnings like :

```
core::num::bignum::Big32x40::mul_small () at src/libcore/num/bignum.rs:254
src/libcore/num/bignum.rs: No such file or directory.
```

That's a known glitch. You can safely ignore those warnings, you're most likely at Reset().

This reset handler will eventually call our main function. Let's skip all the way there using a breakpoint and the `continue` command. To set the breakpoint, let's first take a look where we would like to break in our code, with the `list` command.

```
list main
```

This will show the source code, from the file examples/hello.rs.

```
6       use panic_halt as _;
7
8       use cortex_m_rt::entry;
9       use cortex_m_semihosting::{debug, hprintln};
10
11      #[entry]
12      fn main() -> ! {
13          hprintln!("Hello, world!").unwrap();
14
15          // exit QEMU
```

We would like to add a breakpoint just before the "Hello, world!", which is on line 13. We do that with the `break` command:

```
break 13
```

We can now instruct gdb to run up to our main function, with the `continue` command:

```
continue
```

```
Continuing.

Breakpoint 1, hello::__cortex_m_rt_main () at examples\hello.rs:13
13          hprintln!("Hello, world!").unwrap();
```

We are now close to the code that prints "Hello, world!". Let's move forward using the `next` command.

```
next
```

```
16          debug::exit(debug::EXIT_SUCCESS);
```

At this point you should see "Hello, world!" printed on the terminal that's running `qemu-system-arm`.

```
$ qemu-system-arm (..)
Hello, world!
```

Calling `next` again will terminate the QEMU process.

```
next
```

```
[Inferior 1 (Remote target) exited normally]
```

You can now exit the GDB session.

```
quit
```

Hardware

By now you should be somewhat familiar with the tooling and the development process. In this section we'll switch to real hardware; the process will remain largely the same. Let's dive in.

Know your hardware

Before we begin you need to identify some characteristics of the target device as these will be used to configure the project:

- The ARM core. e.g. Cortex-M3.
- Does the ARM core include an FPU? Cortex-M4**F** and Cortex-M7**F** cores do.
- How much Flash memory and RAM does the target device have? e.g. 256 KiB of Flash and 32 KiB of RAM.
- Where are Flash memory and RAM mapped in the address space? e.g. RAM is commonly located at address `0x2000_0000`.

You can find this information in the data sheet or the reference manual of your device.

In this section we'll be using our reference hardware, the STM32F3DISCOVERY. This board contains an STM32F303VCT6 microcontroller. This microcontroller has:

- A Cortex-M4F core that includes a single precision FPU
- 256 KiB of Flash located at address 0x0800_0000.
- 40 KiB of RAM located at address 0x2000_0000. (There's another RAM region but for simplicity we'll ignore it).

Configuring

We'll start from scratch with a fresh template instance. Refer to the previous section on QEMU for a refresher on how to do this without `cargo-generate`.

```
$ cargo generate --git https://github.com/rust-embedded/cortex-m-quickstart
 Project Name: app
 Creating project called `app`...
 Done! New project created /tmp/app

$ cd app
```

Step number one is to set a default compilation target in `.cargo/config.toml`.

```
tail -n5 .cargo/config.toml


# Pick ONE of these compilation targets
# target = "thumbv6m-none-eabi"    # Cortex-M0 and Cortex-M0+
# target = "thumbv7m-none-eabi"    # Cortex-M3
# target = "thumbv7em-none-eabi"   # Cortex-M4 and Cortex-M7 (no FPU)
target = "thumbv7em-none-eabihf" # Cortex-M4F and Cortex-M7F (with FPU)
```

We'll use `thumbv7em-none-eabihf` as that covers the Cortex-M4F core.

NOTE: As you may remember from the previous chapter, we have to install all targets and this is a new one. So don't forget to run the installation process `rustup target add thumbv7em-none-eabihf` for this target.

The second step is to enter the memory region information into the `memory.x` file.

```
$ cat memory.x
/* Linker script for the STM32F303VCT6 */
MEMORY
{
  /* NOTE 1 K = 1 KiBi = 1024 bytes */
  FLASH : ORIGIN = 0x08000000, LENGTH = 256K
  RAM : ORIGIN = 0x20000000, LENGTH = 40K
}
```

NOTE: If you for some reason changed the `memory.x` file after you had made the first build of a specific build target, then do `cargo clean` before `cargo build`, because `cargo build` may not track updates of `memory.x`.

We'll start with the hello example again, but first we have to make a small change.

In `examples/hello.rs`, make sure the `debug::exit()` call is commented out or removed. It is used only for running in QEMU.

```
#[entry]
fn main() -> ! {
    hprintln!("Hello, world!").unwrap();

    // exit QEMU
    // NOTE do not run this on hardware; it can corrupt OpenOCD state
    // debug::exit(debug::EXIT_SUCCESS);

    loop {}
}
```

You can now cross compile programs using `cargo build` and inspect the binaries using `cargo-binutils` as you did before. The `cortex-m-rt` crate handles all the magic required to get your chip running, as helpfully, pretty much all Cortex-M CPUs boot in the same fashion.

```
cargo build --example hello
```

Debugging

Debugging will look a bit different. In fact, the first steps can look different depending on the target device. In this section we'll show the steps required to debug a program running on the STM32F3DISCOVERY. This is meant to serve as a reference; for device specific information about debugging check out the Debugonomicon.

As before we'll do remote debugging and the client will be a GDB process. This time, however, the server will be OpenOCD.

As done during the verify section connect the discovery board to your laptop / PC and check that the ST-LINK header is populated.

On a terminal run `openocd` to connect to the ST-LINK on the discovery board. Run this command from the root of the template; `openocd` will pick up the `openocd.cfg` file which indicates which interface file and target file to use.

```
cat openocd.cfg
```

```
# Sample OpenOCD configuration for the STM32F3DISCOVERY development board

# Depending on the hardware revision you got you'll have to pick ONE of these
# interfaces. At any time only one interface should be commented out.

# Revision C (newer revision)
source [find interface/stlink.cfg]

# Revision A and B (older revisions)
# source [find interface/stlink-v2.cfg]

source [find target/stm32f3x.cfg]
```

NOTE If you found out that you have an older revision of the discovery board during the verify section then you should modify the `openocd.cfg` file at this point to use `interface/stlink-v2.cfg`.

```
$ openocd
Open On-Chip Debugger 0.10.0
Licensed under GNU GPL v2
For bug reports, read
        http://openocd.org/doc/doxygen/bugs.html
Info : auto-selecting first available session transport "hla_swd". To override use
'transport select <transport>'.
adapter speed: 1000 kHz
adapter_nsrst_delay: 100
Info : The selected transport took over low-level target control. The results
might differ compared to plain JTAG/SWD
none separate
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : Unable to match requested speed 1000 kHz, using 950 kHz
Info : clock speed 950 kHz
Info : STLINK v2 JTAG v27 API v2 SWIM v15 VID 0x0483 PID 0x374B
Info : using stlink api v2
Info : Target voltage: 2.913879
Info : stm32f3x.cpu: hardware has 6 breakpoints, 4 watchpoints
```

On another terminal run GDB, also from the root of the template.

```
gdb-multiarch -q target/thumbv7em-none-eabihf/debug/examples/hello
```

NOTE: like before you might need another version of gdb instead of `gdb-multiarch` depending on which one you installed in the installation chapter. This could also be `arm-none-eabi-gdb` or just `gdb`.

Next connect GDB to OpenOCD, which is waiting for a TCP connection on port 3333.

```
(gdb) target remote :3333
Remote debugging using :3333
0x00000000 in ?? ()
```

Now proceed to *flash* (load) the program onto the microcontroller using the `load` command.

```
(gdb) load
Loading section .vector_table, size 0x400 lma 0x8000000
Loading section .text, size 0x1518 lma 0x8000400
Loading section .rodata, size 0x414 lma 0x8001918
Start address 0x08000400, load size 7468
Transfer rate: 13 KB/sec, 2489 bytes/write.
```

The program is now loaded. This program uses semihosting so before we do any semihosting call we have to tell OpenOCD to enable semihosting. You can send commands to OpenOCD using the `monitor` command.

```
(gdb) monitor arm semihosting enable
semihosting is enabled
```

You can see all the OpenOCD commands by invoking the `monitor help` command.

Like before we can skip all the way to `main` using a breakpoint and the `continue` command.

```
(gdb) break main
Breakpoint 1 at 0x8000490: file examples/hello.rs, line 11.
Note: automatically using hardware breakpoints for read-only addresses.

(gdb) continue
Continuing.

Breakpoint 1, hello::__cortex_m_rt_main_trampoline () at examples/hello.rs:11
11      #[entry]
```

NOTE If GDB blocks the terminal instead of hitting the breakpoint after you issue the `continue` command above, you might want to double check that the memory region information in the `memory.x` file is correctly set up for your device (both the starts *and* lengths).

Step into the main function with `step`.

```
(gdb) step
halted: PC: 0x08000496
hello::__cortex_m_rt_main () at examples/hello.rs:13
13          hprintln!("Hello, world!").unwrap();
```

After advancing the program with `next` you should see "Hello, world!" printed on the OpenOCD console, among other stuff.

```
$ openocd
(..)
Info : halted: PC: 0x08000502
Hello, world!
Info : halted: PC: 0x080004ac
Info : halted: PC: 0x080004ae
Info : halted: PC: 0x080004b0
Info : halted: PC: 0x080004b4
Info : halted: PC: 0x080004b8
Info : halted: PC: 0x080004bc
```

The message is only displayed once as the program is about to enter the infinite loop defined in line 19: `loop {}`

You can now exit GDB using the `quit` command.

```
(gdb) quit
A debugging session is active.

        Inferior 1 [Remote target] will be detached.

Quit anyway? (y or n)
```

Debugging now requires a few more steps so we have packed all those steps into a single GDB script named `openocd.gdb`. The file was created during the `cargo generate` step, and should work without any modifications. Let's have a peek:

```
cat openocd.gdb
```

```
target extended-remote :3333

# print demangled symbols
set print asm-demangle on

# detect unhandled exceptions, hard faults and panics
break DefaultHandler
break HardFault
break rust_begin_unwind

monitor arm semihosting enable

load

# start the process but immediately halt the processor
stepi
```

Now running `<gdb> -x openocd.gdb target/thumbv7em-none-eabihf/debug/examples/hello` will immediately connect GDB to OpenOCD, enable semihosting, load the program and start the process.

Alternatively, you can turn `<gdb> -x openocd.gdb` into a custom runner to make `cargo run` build a program *and* start a GDB session. This runner is included in `.cargo/config.toml` but it's commented out.

```
head -n10 .cargo/config.toml
```

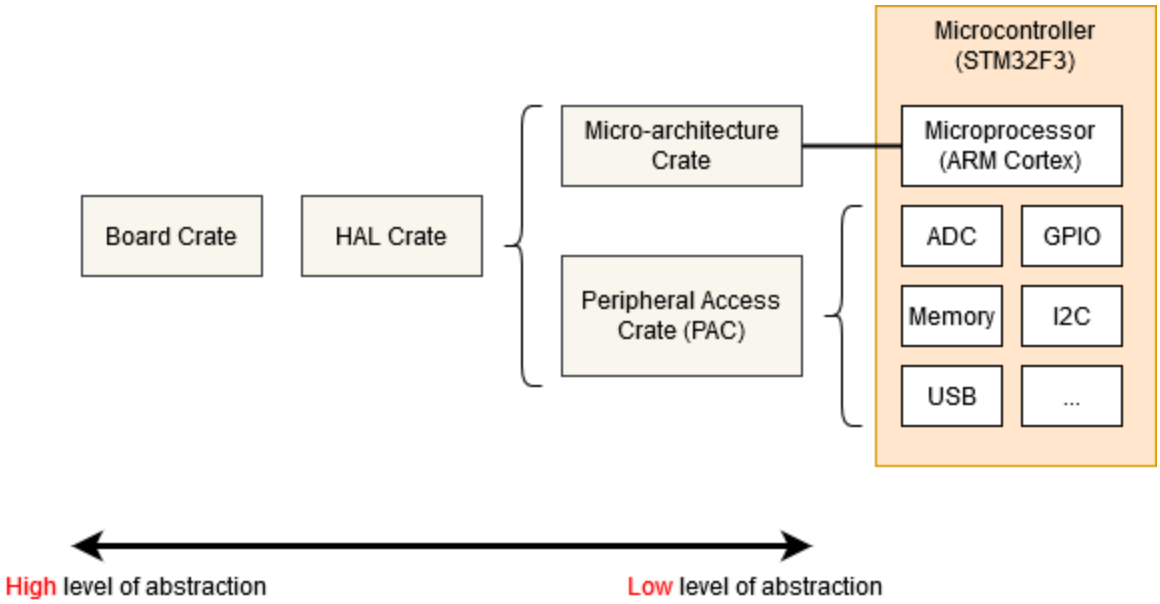
```
[target.thumbv7m-none-eabi]
# uncomment this to make `cargo run` execute programs on QEMU
# runner = "qemu-system-arm -cpu cortex-m3 -machine lm3s6965evb -nographic -
semihosting-config enable=on,target=native -kernel"

[target.'cfg(all(target_arch = "arm", target_os = "none"))']
# uncomment ONE of these three option to make `cargo run` start a GDB session
# which option to pick depends on your system
runner = "arm-none-eabi-gdb -x openocd.gdb"
# runner = "gdb-multiarch -x openocd.gdb"
# runner = "gdb -x openocd.gdb"
```

```
$ cargo run --example hello
(..)
Loading section .vector_table, size 0x400 lma 0x8000000
Loading section .text, size 0x1e70 lma 0x8000400
Loading section .rodata, size 0x61c lma 0x8002270
Start address 0x800144e, load size 10380
Transfer rate: 17 KB/sec, 3460 bytes/write.
(gdb)
```

Memory Mapped Registers

Embedded systems can only get so far by executing normal Rust code and moving data around in RAM. If we want to get any information into or out of our system (be that blinking an LED, detecting a button press or communicating with an off-chip peripheral on some sort of bus) we're going to have to dip into the world of Peripherals and their 'memory mapped registers'.

You may well find that the code you need to access the peripherals in your micro-controller has already been written, at one of the following levels:



- Micro-architecture Crate - This sort of crate handles any useful routines common to the processor core your microcontroller is using, as well as any peripherals that are common to all micro-controllers that use that particular type of processor core. For example the cortex-m crate gives you functions to enable and disable interrupts, which are the same for all Cortex-M based micro-controllers. It also gives you access to the 'SysTick' peripheral included with all Cortex-M based micro-controllers.
- Peripheral Access Crate (PAC) - This sort of crate is a thin wrapper over the various memory-wrapper registers defined for your particular part-number of micro-controller you are using. For example, tm4c123x for the Texas Instruments Tiva-C TM4C123 series, or stm32f30x for the ST-Micro STM32F30x series. Here, you'll be interacting with the registers directly, following each peripheral's operating instructions given in your micro-controller's Technical Reference Manual.
- HAL Crate - These crates offer a more user-friendly API for your particular processor, often by implementing some common traits defined in embedded-hal. For example, this crate might offer a `Serial` struct, with a constructor that takes an appropriate set of GPIO pins and a baud rate, and offers some sort of `write_byte` function for sending data. See the chapter on Portability for more information on embedded-hal.

- Board Crate - These crates go one step further than a HAL Crate by pre-configuring various peripherals and GPIO pins to suit the specific developer kit or board you are using, such as stm32f3-discovery for the STM32F3DISCOVERY board.

Board Crate

A board crate is the perfect starting point, if you're new to embedded Rust. They nicely abstract the HW details that might be overwhelming when starting studying this subject, and makes standard tasks easy, like turning a LED on or off. The functionality it exposes varies a lot between boards. Since this book aims at staying hardware agnostic, the board crates won't be covered by this book.

If you want to experiment with the STM32F3DISCOVERY board, it is highly recommended to take a look at the stm32f3-discovery board crate, which provides functionality to blink the board LEDs, access its compass, bluetooth and more. The Discovery book offers a great introduction to the use of a board crate.

But if you're working on a system that doesn't yet have dedicated board crate, or you need functionality not provided by existing crates, read on as we start from the bottom, with the micro-architecture crates.

Micro-architecture crate

Let's look at the SysTick peripheral that's common to all Cortex-M based micro-controllers. We can find a pretty low-level API in the cortex-m crate, and we can use it like this:

```rust
#![no_std]
#![no_main]
use cortex_m::peripheral::{syst, Peripherals};
use cortex_m_rt::entry;
use panic_halt as _;

#[entry]
fn main() -> ! {
    let peripherals = Peripherals::take().unwrap();
    let mut systick = peripherals.SYST;
    systick.set_clock_source(syst::SystClkSource::Core);
    systick.set_reload(1_000);
    systick.clear_current();
    systick.enable_counter();
    while !systick.has_wrapped() {
        // Loop
    }

    loop {}
}
```

The functions on the `SYST` struct map pretty closely to the functionality defined by the ARM Technical Reference Manual for this peripheral. There's nothing in this API about 'delaying for X milliseconds' - we have to crudely implement that ourselves using a `while` loop. Note that we can't access our `SYST` struct until we have called `Peripherals::take()` - this is a special routine that guarantees that there is only one `SYST` structure in our entire program. For more on that, see the Peripherals section.

Using a Peripheral Access Crate (PAC)

We won't get very far with our embedded software development if we restrict ourselves to only the basic peripherals included with every Cortex-M. At some point, we're going to need to write some code that's specific to the particular micro-controller we're using. In this example, let's assume we have an Texas Instruments TM4C123 - a middling 80MHz Cortex-M4 with 256 KiB of Flash. We're going to pull in the tm4c123x crate to make use of this chip.

```
#![no_std]
#![no_main]

use panic_halt as _; // panic handler

use cortex_m_rt::entry;
use tm4c123x;

#[entry]
pub fn init() -> (Delay, Leds) {
    let cp = cortex_m::Peripherals::take().unwrap();
    let p = tm4c123x::Peripherals::take().unwrap();

    let pwm = p.PWM0;
    pwm.ctl.write(|w| w.globalsync0().clear_bit());
    // Mode = 1 => Count up/down mode
    pwm._2_ctl.write(|w| w.enable().set_bit().mode().set_bit());
    pwm._2_gena.write(|w| w.actcmpau().zero().actcmpad().one());
    // 528 cycles (264 up and down) = 4 loops per video line (2112 cycles)
    pwm._2_load.write(|w| unsafe { w.load().bits(263) });
    pwm._2_cmpa.write(|w| unsafe { w.compa().bits(64) });
    pwm.enable.write(|w| w.pwm4en().set_bit());
}
```

We've accessed the `PWM0` peripheral in exactly the same way as we accessed the `SYST` peripheral earlier, except we called `tm4c123x::Peripherals::take()`. As this crate was auto-generated using svd2rust, the access functions for our register fields take a closure, rather than a numeric argument. While this looks like a lot of code, the Rust compiler can use it to perform a bunch of checks for us, but then generate machine-code which is pretty close to hand-written assembler! Where the auto-generated code isn't able to determine that all possible arguments to a particular accessor function are valid (for example, if the SVD defines the register as 32-bit but doesn't say if some of those 32-bit values have a special meaning), then the function is marked as `unsafe`. We can see this in the example above when setting the `load` and `compa` sub-fields using the `bits()` function.

Reading

The `read()` function returns an object which gives read-only access to the various sub-fields within this register, as defined by the manufacturer's SVD file for this chip. You can find all the functions available on special `R` return type for this particular register, in this particular peripheral, on this particular chip, in the tm4c123x documentation.

```
if pwm.ctl.read().globalsync0().is_set() {
    // Do a thing
}
```

Writing

The `write()` function takes a closure with a single argument. Typically we call this `w`. This argument then gives read-write access to the various sub-fields within this register, as defined by the manufacturer's SVD file for this chip. Again, you can find all the functions available on the 'w' for this particular register, in this particular peripheral, on this particular chip, in the tm4c123x documentation. Note that all of the sub-fields that we do not set will be set to a default value for us - any existing content in the register will be lost.

```
pwm.ctl.write(|w| w.globalsync0().clear_bit());
```

Modifying

If we wish to change only one particular sub-field in this register and leave the other sub-fields unchanged, we can use the `modify` function. This function takes a closure with two arguments - one for reading and one for writing. Typically we call these `r` and `w` respectively. The `r` argument can be used to inspect the current contents of the register, and the `w` argument can be used to modify the register contents.

```
pwm.ctl.modify(|r, w| w.globalsync0().clear_bit());
```

The `modify` function really shows the power of closures here. In C, we'd have to read into some temporary value, modify the correct bits and then write the value back. This means there's considerable scope for error:

```
uint32_t temp = pwm0.ctl.read();
temp |= PWM0_CTL_GLOBALSYNC0;
pwm0.ctl.write(temp);
uint32_t temp2 = pwm0.enable.read();
temp2 |= PWM0_ENABLE_PWM4EN;
pwm0.enable.write(temp); // Uh oh! Wrong variable!
```

Using a HAL crate

The HAL crate for a chip typically works by implementing a custom Trait for the raw structures exposed by the PAC. Often this trait will define a function called `constrain()` for single peripherals or `split()` for things like GPIO ports with multiple pins. This function will consume the underlying raw peripheral structure and return a new object with a higher-level API. This API may also do things like have the Serial port `new` function require a borrow on some `Clock`

structure, which can only be generated by calling the function which configures the PLLs and sets up all the clock frequencies. In this way, it is statically impossible to create a Serial port object without first having configured the clock rates, or for the Serial port object to misconvert the baud rate into clock ticks. Some crates even define special traits for the states each GPIO pin can be in, requiring the user to put a pin into the correct state (say, by selecting the appropriate Alternate Function Mode) before passing the pin into Peripheral. All with no run-time cost!

Let's see an example:

```rust
#![no_std]
#![no_main]

use panic_halt as _; // panic handler

use cortex_m_rt::entry;
use tm4c123x_hal as hal;
use tm4c123x_hal::prelude::*;
use tm4c123x_hal::serial::{NewlineMode, Serial};
use tm4c123x_hal::sysctl;

#[entry]
fn main() -> ! {
    let p = hal::Peripherals::take().unwrap();
    let cp = hal::CorePeripherals::take().unwrap();

    // Wrap up the SYSCTL struct into an object with a higher-layer API
    let mut sc = p.SYSCTL.constrain();
    // Pick our oscillation settings
    sc.clock_setup.oscillator = sysctl::Oscillator::Main(
        sysctl::CrystalFrequency::_16mhz,
        sysctl::SystemClock::UsePll(sysctl::PllOutputFrequency::_80_00mhz),
    );
    // Configure the PLL with those settings
    let clocks = sc.clock_setup.freeze();

    // Wrap up the GPIO_PORTA struct into an object with a higher-layer API.
    // Note it needs to borrow `sc.power_control` so it can power up the GPIO
    // peripheral automatically.
    let mut porta = p.GPIO_PORTA.split(&sc.power_control);

    // Activate the UART.
    let uart = Serial::uart0(
        p.UART0,
        // The transmit pin
        porta
            .pa1
            .into_af_push_pull::<hal::gpio::AF1>(&mut porta.control),
        // The receive pin
        porta
            .pa0
            .into_af_push_pull::<hal::gpio::AF1>(&mut porta.control),
        // No RTS or CTS required
        (),
        (),
        // The baud rate
        115200_u32.bps(),
        // Output handling
        NewlineMode::SwapLFtoCRLF,
        // We need the clock rates to calculate the baud rate divisors
        &clocks,
        // We need this to power up the UART peripheral
        &sc.power_control,
```

```
        );

        loop {
            writeln!(uart, "Hello, World!\r\n").unwrap();
        }
    }
```

Semihosting

Semihosting is a mechanism that lets embedded devices do I/O on the host and is mainly used to log messages to the host console. Semihosting requires a debug session and pretty much nothing else (no extra wires!) so it's super convenient to use. The downside is that it's super slow: each write operation can take several milliseconds depending on the hardware debugger (e.g. ST-Link) you use.

The `cortex-m-semihosting` crate provides an API to do semihosting operations on Cortex-M devices. The program below is the semihosting version of "Hello, world!":

```
#![no_main]
#![no_std]

use panic_halt as _;

use cortex_m_rt::entry;
use cortex_m_semihosting::hprintln;

#[entry]
fn main() -> ! {
    hprintln!("Hello, world!").unwrap();

    loop {}
}
```

If you run this program on hardware you'll see the "Hello, world!" message within the OpenOCD logs.

```
$ openocd
(..)
Hello, world!
(..)
```

You do need to enable semihosting in OpenOCD from GDB first:

```
(gdb) monitor arm semihosting enable
semihosting is enabled
```

QEMU understands semihosting operations so the above program will also work with `qemu-system-arm` without having to start a debug session. Note that you'll need to pass the `-semihosting-config` flag to QEMU to enable semihosting support; these flags are already included in the `.cargo/config.toml` file of the template.

```
$ # this program will block the terminal
$ cargo run
     Running `qemu-system-arm (..)
Hello, world!
```

There's also an `exit` semihosting operation that can be used to terminate the QEMU process. Important: do **not** use `debug::exit` on hardware; this function can corrupt your OpenOCD session and you will not be able to debug more programs until you restart it.

```
#![no_main]
#![no_std]

use panic_halt as _;

use cortex_m_rt::entry;
use cortex_m_semihosting::debug;

#[entry]
fn main() -> ! {
    let roses = "blue";

    if roses == "red" {
        debug::exit(debug::EXIT_SUCCESS);
    } else {
        debug::exit(debug::EXIT_FAILURE);
    }

    loop {}
}
```

```
$ cargo run
     Running `qemu-system-arm (..)

$ echo $?
1
```

One last tip: you can set the panicking behavior to `exit(EXIT_FAILURE)`. This will let you write `no_std` run-pass tests that you can run on QEMU.

For convenience, the `panic-semihosting` crate has an "exit" feature that when enabled invokes `exit(EXIT_FAILURE)` after logging the panic message to the host stderr.

```rust
#![no_main]
#![no_std]

use panic_semihosting as _; // features = ["exit"]

use cortex_m_rt::entry;
use cortex_m_semihosting::debug;

#[entry]
fn main() -> ! {
    let roses = "blue";

    assert_eq!(roses, "red");

    loop {}
}
```

```
$ cargo run
     Running `qemu-system-arm (..)
panicked at 'assertion failed: `(left == right)`
  left: `"blue"`,
 right: `"red"`', examples/hello.rs:15:5

$ echo $?
1
```

NOTE: To enable this feature on `panic-semihosting`, edit your `Cargo.toml` dependencies section where `panic-semihosting` is specified with:

```
panic-semihosting = { version = "VERSION", features = ["exit"] }
```

where `VERSION` is the version desired. For more information on dependencies features check the `specifying dependencies` section of the Cargo book.

Panicking

Panicking is a core part of the Rust language. Built-in operations like indexing are runtime checked for memory safety. When out of bounds indexing is attempted this results in a panic.

In the standard library panicking has a defined behavior: it unwinds the stack of the panicking thread, unless the user opted for aborting the program on panics.

In programs without standard library, however, the panicking behavior is left undefined. A behavior can be chosen by declaring a `#[panic_handler]` function. This function must appear exactly *once* in the dependency graph of a program, and must have the following signature: `fn(&PanicInfo) -> !`, where `PanicInfo` is a struct containing information about the location of the panic.

Given that embedded systems range from user facing to safety critical (cannot crash) there's no one size fits all panicking behavior but there are plenty of commonly used behaviors. These common behaviors have been packaged into crates that define the `#[panic_handler]` function. Some examples include:

- `panic-abort`. A panic causes the abort instruction to be executed.
- `panic-halt`. A panic causes the program, or the current thread, to halt by entering an infinite loop.
- `panic-itm`. The panicking message is logged using the ITM, an ARM Cortex-M specific peripheral.
- `panic-semihosting`. The panicking message is logged to the host using the semihosting technique.

You may be able to find even more crates searching for the `panic-handler` keyword on crates.io.

A program can pick one of these behaviors simply by linking to the corresponding crate. The fact that the panicking behavior is expressed in the source of an application as a single line of code is not only useful as documentation but can also be used to change the panicking behavior according to the compilation profile. For example:

```
#![no_main]
#![no_std]

// dev profile: easier to debug panics; can put a breakpoint on
`rust_begin_unwind`
#[cfg(debug_assertions)]
use panic_halt as _;

// release profile: minimize the binary size of the application
#[cfg(not(debug_assertions))]
use panic_abort as _;

// ..
```

In this example the crate links to the `panic-halt` crate when built with the dev profile (`cargo build`), but links to the `panic-abort` crate when built with the release profile (`cargo build --release`).

> The `use panic_abort as _;` form of the `use` statement is used to ensure the `panic_abort` panic handler is included in our final executable while making it clear to the compiler that we won't explicitly use anything from the crate. Without the `as _` rename, the compiler would warn that we have an unused import. Sometimes you might see `extern crate panic_abort` instead, which is an older style used before the 2018 edition of Rust, and should now only be used for "sysroot" crates (those distributed with Rust itself) such as `proc_macro` , `alloc` , `std` , and `test` .

An example

Here's an example that tries to index an array beyond its length. The operation results in a panic.

```rust
#![no_main]
#![no_std]

use panic_semihosting as _;

use cortex_m_rt::entry;

#[entry]
fn main() -> ! {
    let xs = [0, 1, 2];
    let i = xs.len();
    let _y = xs[i]; // out of bounds access

    loop {}
}
```

This example chose the `panic-semihosting` behavior which prints the panic message to the host console using semihosting.

```
$ cargo run
     Running `qemu-system-arm -cpu cortex-m3 -machine lm3s6965evb (..)
panicked at 'index out of bounds: the len is 3 but the index is 4',
src/main.rs:12:13
```

You can try changing the behavior to `panic-halt` and confirm that no message is printed in that case.

Exceptions

Exceptions, and interrupts, are a hardware mechanism by which the processor handles asynchronous events and fatal errors (e.g. executing an invalid instruction). Exceptions imply preemption and involve exception handlers, subroutines executed in response to the signal that triggered the event.

The `cortex-m-rt` crate provides an `exception` attribute to declare exception handlers.

```
// Exception handler for the SysTick (System Timer) exception
#[exception]
fn SysTick() {
    // ..
}
```

Other than the `exception` attribute exception handlers look like plain functions but there's one more difference: `exception` handlers can *not* be called by software. Following the previous example, the statement `SysTick();` would result in a compilation error.

This behavior is pretty much intended and it's required to provide a feature: `static mut` variables declared *inside* `exception` handlers are *safe* to use.

```
#[exception]
fn SysTick() {
    static mut COUNT: u32 = 0;

    // `COUNT` has transformed to type `&mut u32` and it's safe to use
    *COUNT += 1;
}
```

As you may know, using `static mut` variables in a function makes it *non-reentrant*. It's undefined behavior to call a non-reentrant function, directly or indirectly, from more than one exception / interrupt handler or from `main` and one or more exception / interrupt handlers.

Safe Rust must never result in undefined behavior so non-reentrant functions must be marked as `unsafe`. Yet I just told that `exception` handlers can safely use `static mut` variables. How is this possible? This is possible because `exception` handlers can *not* be called by software thus reentrancy is not possible. These handlers are called by the hardware itself which is assumed to be physically non-concurrent.

As a result, in the context of exception handlers in embedded systems, the absence of concurrent invocations of the same handler ensures that there are no reentrancy issues, even if the handler uses static mutable variables.

In a multicore system, where multiple processor cores are executing code concurrently, the potential for reentrancy issues becomes relevant again, even within exception handlers. While each core may have its own set of exception handlers, there can still be scenarios where multiple cores attempt to execute the same exception handler simultaneously.
To address this concern in a multicore environment, proper synchronization mechanisms need to be employed within the exception handlers to ensure that access to shared resources is properly coordinated among the cores. This typically involves the use of techniques such as locks, semaphores, or atomic operations to prevent data races and maintain data integrity

---

Note that the `exception` attribute transforms definitions of static variables inside the function by wrapping them into `unsafe` blocks and providing us with new appropriate variables of type `&mut` of the same name. Thus we can dereference the reference via `*` to access the values of the variables without needing to wrap them in an `unsafe` block.

A complete example

Here's an example that uses the system timer to raise a `SysTick` exception roughly every second. The `SysTick` exception handler keeps track of how many times it has been called in the `COUNT` variable and then prints the value of `COUNT` to the host console using semihosting.

NOTE: You can run this example on any Cortex-M device; you can also run it on QEMU

```rust
#![deny(unsafe_code)]
#![no_main]
#![no_std]

use panic_halt as _;

use core::fmt::Write;

use cortex_m::peripheral::syst::SystClkSource;
use cortex_m_rt::{entry, exception};
use cortex_m_semihosting::{
    debug,
    hio::{self, HStdout},
};

#[entry]
fn main() -> ! {
    let p = cortex_m::Peripherals::take().unwrap();
    let mut syst = p.SYST;

    // configures the system timer to trigger a SysTick exception every second
    syst.set_clock_source(SystClkSource::Core);
    // this is configured for the LM3S6965 which has a default CPU clock of 12 MHz
    syst.set_reload(12_000_000);
    syst.clear_current();
    syst.enable_counter();
    syst.enable_interrupt();

    loop {}
}

#[exception]
fn SysTick() {
    static mut COUNT: u32 = 0;
    static mut STDOUT: Option<HStdout> = None;

    *COUNT += 1;

    // Lazy initialization
    if STDOUT.is_none() {
        *STDOUT = hio::hstdout().ok();
    }

    if let Some(hstdout) = STDOUT.as_mut() {
        write!(hstdout, "{}", *COUNT).ok();
    }

    // IMPORTANT omit this `if` block if running on real hardware or your
    // debugger will end in an inconsistent state
    if *COUNT == 9 {
        // This will terminate the QEMU process
        debug::exit(debug::EXIT_SUCCESS);
```

```
    }
}
```

```
tail -n5 Cargo.toml
```

```
[dependencies]
cortex-m = "0.5.7"
cortex-m-rt = "0.6.3"
panic-halt = "0.2.0"
cortex-m-semihosting = "0.3.1"
```

```
$ cargo run --release
     Running `qemu-system-arm -cpu cortex-m3 -machine lm3s6965evb (..)
123456789
```

If you run this on the Discovery board you'll see the output on the OpenOCD console. Also, the program will *not* stop when the count reaches 9.

The default exception handler

What the `exception` attribute actually does is *override* the default exception handler for a specific exception. If you don't override the handler for a particular exception it will be handled by the `DefaultHandler` function, which defaults to:

```
fn DefaultHandler() {
    loop {}
}
```

This function is provided by the `cortex-m-rt` crate and marked as `#[no_mangle]` so you can put a breakpoint on "DefaultHandler" and catch *unhandled* exceptions.

It's possible to override this `DefaultHandler` using the `exception` attribute:

```
#[exception]
fn DefaultHandler(irqn: i16) {
    // custom default handler
}
```

The `irqn` argument indicates which exception is being serviced. A negative value indicates that a Cortex-M exception is being serviced; and zero or a positive value indicate that a device specific exception, AKA interrupt, is being serviced.

The hard fault handler

The `HardFault` exception is a bit special. This exception is fired when the program enters an invalid state so its handler can *not* return as that could result in undefined behavior. Also, the runtime crate does a bit of work before the user defined `HardFault` handler is invoked to improve debuggability.

The result is that the `HardFault` handler must have the following signature: `fn(&ExceptionFrame) -> !`. The argument of the handler is a pointer to registers that were pushed into the stack by the exception. These registers are a snapshot of the processor state at the moment the exception was triggered and are useful to diagnose a hard fault.

Here's an example that performs an illegal operation: a read to a nonexistent memory location.

> **NOTE**: This program won't work, i.e. it won't crash, on QEMU because `qemu-system-arm -machine lm3s6965evb` doesn't check memory loads and will happily return `0` on reads to invalid memory.

```rust
#![no_main]
#![no_std]

use panic_halt as _;

use core::fmt::Write;
use core::ptr;

use cortex_m_rt::{entry, exception, ExceptionFrame};
use cortex_m_semihosting::hio;

#[entry]
fn main() -> ! {
    // read a nonexistent memory location
    unsafe {
        ptr::read_volatile(0x3FFF_0000 as *const u32);
    }

    loop {}
}

#[exception]
fn HardFault(ef: &ExceptionFrame) -> ! {
    if let Ok(mut hstdout) = hio::hstdout() {
        writeln!(hstdout, "{:#?}", ef).ok();
    }

    loop {}
}
```

The `HardFault` handler prints the `ExceptionFrame` value. If you run this you'll see something like this on the OpenOCD console.

```
$ openocd
(..)
ExceptionFrame {
    r0: 0x3fff0000,
    r1: 0x00000003,
    r2: 0x080032e8,
    r3: 0x00000000,
    r12: 0x00000000,
    lr: 0x080016df,
    pc: 0x080016e2,
    xpsr: 0x61000000,
}
```

The `pc` value is the value of the Program Counter at the time of the exception and it points to the instruction that triggered the exception.

If you look at the disassembly of the program:

```
$ cargo objdump --bin app --release -- -d --no-show-raw-insn --print-imm-hex
(..)
ResetTrampoline:
 8000942:       movw    r0, #0xfffe
 8000946:       movt    r0, #0x3fff
 800094a:       ldr     r0, [r0]
 800094c:       b       #-0x4 <ResetTrampoline+0xa>
```

You can lookup the value of the program counter `0x0800094a` in the disassembly. You'll see that a load operation (`ldr r0, [r0]`) caused the exception. The `r0` field of `ExceptionFrame` will tell you the value of register `r0` was `0x3fff_fffe` at that time.

Interrupts

Interrupts differ from exceptions in a variety of ways but their operation and use is largely similar and they are also handled by the same interrupt controller. Whereas exceptions are defined by the Cortex-M architecture, interrupts are always vendor (and often even chip) specific implementations, both in naming and functionality.

Interrupts do allow for a lot of flexibility which needs to be accounted for when attempting to use them in an advanced way. We will not cover those uses in this book, however it is a good idea to keep the following in mind:

- Interrupts have programmable priorities which determine their handlers' execution order
- Interrupts can nest and preempt, i.e. execution of an interrupt handler might be interrupted by another higher-priority interrupt
- In general the reason causing the interrupt to trigger needs to be cleared to prevent re-entering the interrupt handler endlessly

The general initialization steps at runtime are always the same:

- Setup the peripheral(s) to generate interrupts requests at the desired occasions
- Set the desired priority of the interrupt handler in the interrupt controller
- Enable the interrupt handler in the interrupt controller

Similarly to exceptions, the `cortex-m-rt` crate provides an `interrupt` attribute to declare interrupt handlers. The available interrupts (and their position in the interrupt handler table) are usually automatically generated via `svd2rust` from a SVD description.

```
// Interrupt handler for the Timer2 interrupt
#[interrupt]
fn TIM2() {
    // ..
    // Clear reason for the generated interrupt request
}
```

Interrupt handlers look like plain functions (except for the lack of arguments) similar to exception handlers. However they can not be called directly by other parts of the firmware due to the special calling conventions. It is however possible to generate interrupt requests in software to trigger a diversion to the interrupt handler.

Similar to exception handlers it is also possible to declare `static mut` variables inside the interrupt handlers for *safe* state keeping.

```
#[interrupt]
fn TIM2() {
    static mut COUNT: u32 = 0;

    // `COUNT` has type `&mut u32` and it's safe to use
    *COUNT += 1;
}
```

For a more detailed description about the mechanisms demonstrated here please refer to the exceptions section.

IO

TODO Cover memory mapped I/O using registers.

Peripherals

What are Peripherals?

Most Microcontrollers have more than just a CPU, RAM, or Flash Memory - they contain sections of silicon which are used for interacting with systems outside of the microcontroller, as well as directly and indirectly interacting with their surroundings in the world via sensors, motor controllers, or human interfaces such as a display or keyboard. These components are collectively known as Peripherals.

These peripherals are useful because they allow a developer to offload processing to them, avoiding having to handle everything in software. Similar to how a desktop developer would offload graphics processing to a video card, embedded developers can offload some tasks to peripherals allowing the CPU to spend its time doing something else important, or doing nothing in order to save power.

If you look at the main circuit board in an old-fashioned home computer from the 1970s or 1980s (and actually, the desktop PCs of yesterday are not so far removed from the embedded systems of today) you would expect to see:

- A processor
- A RAM chip
- A ROM chip
- An I/O controller

The RAM chip, ROM chip and I/O controller (the peripheral in this system) would be joined to the processor through a series of parallel traces known as a 'bus'. This bus carries address information, which selects which device on the bus the processor wishes to communicate with, and a data bus which carries the actual data. In our embedded microcontrollers, the same principles apply - it's just that everything is packed on to a single piece of silicon.

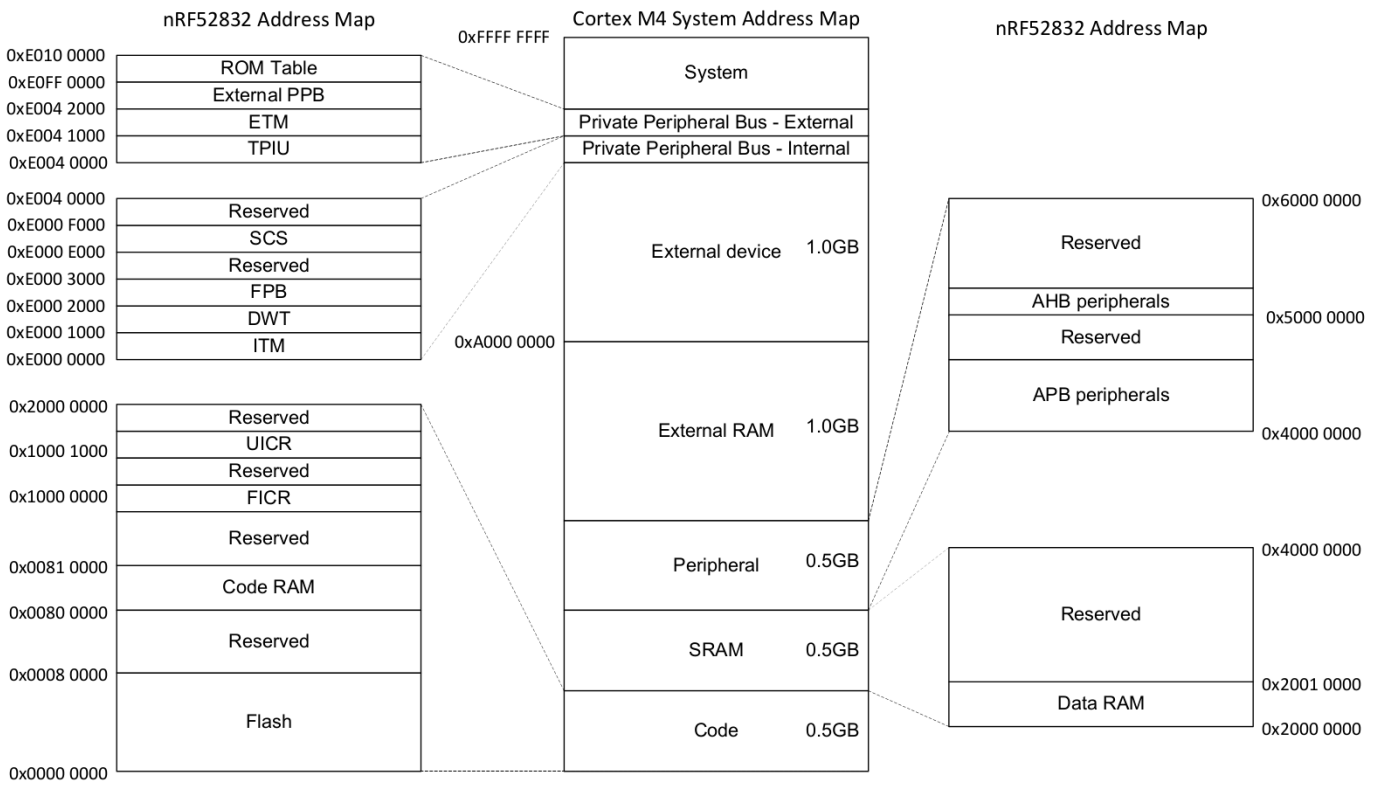
However, unlike graphics cards, which typically have a Software API like Vulkan, Metal, or OpenGL, peripherals are exposed to our Microcontroller with a hardware interface, which is mapped to a chunk of the memory.

Linear and Real Memory Space

On a microcontroller, writing some data to some other arbitrary address, such as `0x4000_0000` or `0x0000_0000`, may also be a completely valid action.

On a desktop system, access to memory is tightly controlled by the MMU, or Memory Management Unit. This component has two major responsibilities: enforcing access permission to sections of memory (preventing one process from reading or modifying the memory of another process); and re-mapping segments of the physical memory to virtual memory ranges used in software. Microcontrollers do not typically have an MMU, and instead only use real physical addresses in software.

Although 32 bit microcontrollers have a real and linear address space from `0x0000_0000`, and `0xFFFF_FFFF`, they generally only use a few hundred kilobytes of that range for actual memory. This leaves a significant amount of address space remaining. In earlier chapters, we were talking about RAM being located at address `0x2000_0000`. If our RAM was 64 KiB long (i.e. with a maximum address of 0xFFFF) then addresses `0x2000_0000` to `0x2000_FFFF` would correspond to our RAM. When we write to a variable which lives at address `0x2000_1234`, what happens internally is that some logic detects the upper portion of the address (0x2000 in this example) and then activates the RAM so that it can act upon the lower portion of the address (0x1234 in this case). On a Cortex-M we also have our Flash ROM mapped in at address `0x0000_0000` up to, say, address `0x0007_FFFF` (if we have a 512 KiB Flash ROM). Rather than ignore all remaining space between these two regions, Microcontroller designers instead mapped the interface for peripherals in certain memory locations. This ends up looking something like this:



Nordic nRF52832 Datasheet (pdf)

Memory Mapped Peripherals

Interaction with these peripherals is simple at a first glance - write the right data to the correct address. For example, sending a 32 bit word over a serial port could be as direct as writing that 32 bit word to a certain memory address. The Serial Port Peripheral would then take over and send out the data automatically.

Configuration of these peripherals works similarly. Instead of calling a function to configure a peripheral, a chunk of memory is exposed which serves as the hardware API. Write `0x8000_0000` to a SPI Frequency Configuration Register, and the SPI port will send data at 8 Megabits per second. Write `0x0200_0000` to the same address, and the SPI port will send data at 125 Kilobits per second. These configuration registers look a little bit like this:

31.6.8 FREQUENCY

Address offset: 0x524

SPI frequency

Bit number				31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Id				A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A	A
Reset 0x04000000				**0**	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Id	**RW**	**Field**	**Value Id**	**Value**								**Description**																							
A	RW	FREQUENCY										SPI master data rate																							
			K125	0x02000000								125 kbps																							
			K250	0x04000000								250 kbps																							
			K500	0x08000000								500 kbps																							
			M1	0x10000000								1 Mbps																							
			M2	0x20000000								2 Mbps																							
			M4	0x40000000								4 Mbps																							
			M8	0x80000000								8 Mbps																							

Nordic nRF52832 Datasheet (pdf)

This interface is how interactions with the hardware are made, no matter what language is used, whether that language is Assembly, C, or Rust.

A First Attempt

The Registers

Let's look at the 'SysTick' peripheral - a simple timer which comes with every Cortex-M processor core. Typically you'll be looking these up in the chip manufacturer's data sheet or *Technical Reference Manual*, but this example is common to all ARM Cortex-M cores, let's look in the ARM reference manual. We see there are four registers:

Offset	Name	Description	Width
0x00	SYST_CSR	Control and Status Register	32 bits
0x04	SYST_RVR	Reload Value Register	32 bits
0x08	SYST_CVR	Current Value Register	32 bits
0x0C	SYST_CALIB	Calibration Value Register	32 bits

The C Approach

In Rust, we can represent a collection of registers in exactly the same way as we do in C - with a `struct`.

```
#[repr(C)]
struct SysTick {
    pub csr: u32,
    pub rvr: u32,
    pub cvr: u32,
    pub calib: u32,
}
```

The qualifier `#[repr(C)]` tells the Rust compiler to lay this structure out like a C compiler would. That's very important, as Rust allows structure fields to be re-ordered, while C does not. You can imagine the debugging we'd have to do if these fields were silently re-arranged by the compiler! With this qualifier in place, we have our four 32-bit fields which correspond to the table above. But of course, this `struct` is of no use by itself - we need a variable.

```
let systick = 0xE000_E010 as *mut SysTick;
let time = unsafe { (*systick).cvr };
```

Volatile Accesses

Now, there are a couple of problems with the approach above.

1. We have to use unsafe every time we want to access our Peripheral.
2. We've got no way of specifying which registers are read-only or read-write.
3. Any piece of code anywhere in your program could access the hardware through this structure.
4. Most importantly, it doesn't actually work...

Now, the problem is that compilers are clever. If you make two writes to the same piece of RAM, one after the other, the compiler can notice this and just skip the first write entirely. In C, we can mark variables as `volatile` to ensure that every read or write occurs as intended. In Rust, we instead mark the *accesses* as volatile, not the variable.

```
let systick = unsafe { &mut *(0xE000_E010 as *mut SysTick) };
let time = unsafe { core::ptr::read_volatile(&mut systick.cvr) };
```

So, we've fixed one of our four problems, but now we have even more `unsafe` code! Fortunately, there's a third party crate which can help - `volatile_register`.

```
use volatile_register::{RW, RO};

#[repr(C)]
struct SysTick {
    pub csr: RW<u32>,
    pub rvr: RW<u32>,
    pub cvr: RW<u32>,
    pub calib: RO<u32>,
}

fn get_systick() -> &'static mut SysTick {
    unsafe { &mut *(0xE000_E010 as *mut SysTick) }
}

fn get_time() -> u32 {
    let systick = get_systick();
    systick.cvr.read()
}
```

Now, the volatile accesses are performed automatically through the `read` and `write` methods. It's still `unsafe` to perform writes, but to be fair, hardware is a bunch of mutable state and there's no way for the compiler to know whether these writes are actually safe, so this is a good default position.

The Rusty Wrapper

We need to wrap this `struct` up into a higher-layer API that is safe for our users to call. As the driver author, we manually verify the unsafe code is correct, and then present a safe API for our users so they don't have to worry about it (provided they trust us to get it right!).

One example might be:

```
use volatile_register::{RW, RO};

pub struct SystemTimer {
    p: &'static mut RegisterBlock
}

#[repr(C)]
struct RegisterBlock {
    pub csr: RW<u32>,
    pub rvr: RW<u32>,
    pub cvr: RW<u32>,
    pub calib: RO<u32>,
}

impl SystemTimer {
    pub fn new() -> SystemTimer {
        SystemTimer {
            p: unsafe { &mut *(0xE000_E010 as *mut RegisterBlock) }
        }
    }

    pub fn get_time(&self) -> u32 {
        self.p.cvr.read()
    }

    pub fn set_reload(&mut self, reload_value: u32) {
        unsafe { self.p.rvr.write(reload_value) }
    }
}

pub fn example_usage() -> String {
    let mut st = SystemTimer::new();
    st.set_reload(0x00FF_FFFF);
    format!("Time is now 0x{:08x}", st.get_time())
}
```

Now, the problem with this approach is that the following code is perfectly acceptable to the compiler:

```
fn thread1() {
    let mut st = SystemTimer::new();
    st.set_reload(2000);
}

fn thread2() {
    let mut st = SystemTimer::new();
    st.set_reload(1000);
}
```

Our `&mut self` argument to the `set_reload` function checks that there are no other references to *that* particular `SystemTimer` struct, but they don't stop the user creating a second `SystemTimer` which points to the exact same peripheral! Code written in this fashion will work if the author is diligent enough to spot all of these 'duplicate' driver instances, but once the code is spread out over multiple modules, drivers, developers, and days, it gets easier and easier to make these kinds of mistakes.

Mutable Global State

Unfortunately, hardware is basically nothing but mutable global state, which can feel very frightening for a Rust developer. Hardware exists independently from the structures of the code we write, and can be modified at any time by the real world.

What should our rules be?

How can we reliably interact with these peripherals?

1. Always use `volatile` methods to read or write to peripheral memory, as it can change at any time
2. In software, we should be able to share any number of read-only accesses to these peripherals
3. If some software should have read-write access to a peripheral, it should hold the only reference to that peripheral

The Borrow Checker

The last two of these rules sound suspiciously similar to what the Borrow Checker does already!

Imagine if we could pass around ownership of these peripherals, or offer immutable or mutable references to them?

Well, we can, but for the Borrow Checker, we need to have exactly one instance of each peripheral, so Rust can handle this correctly. Well, luckily in the hardware, there is only one instance of any given peripheral, but how can we expose that in the structure of our code?

Singletons

> In software engineering, the singleton pattern is a software design pattern that restricts the instantiation of a class to one object.
>
> *Wikipedia: Singleton Pattern*

But why can't we just use global variable(s)?

We could make everything a public static, like this

```
static mut THE_SERIAL_PORT: SerialPort = SerialPort;

fn main() {
    let _ = unsafe {
        THE_SERIAL_PORT.read_speed();
    };
}
```

But this has a few problems. It is a mutable global variable, and in Rust, these are always unsafe to interact with. These variables are also visible across your whole program, which means the borrow checker is unable to help you track references and ownership of these variables.

How do we do this in Rust?

Instead of just making our peripheral a global variable, we might instead decide to make a structure, in this case called `PERIPHERALS`, which contains an `Option<T>` for each of our peripherals.

```
struct Peripherals {
    serial: Option<SerialPort>,
}
impl Peripherals {
    fn take_serial(&mut self) -> SerialPort {
        let p = replace(&mut self.serial, None);
        p.unwrap()
    }
}
static mut PERIPHERALS: Peripherals = Peripherals {
    serial: Some(SerialPort),
};
```

This structure allows us to obtain a single instance of our peripheral. If we try to call `take_serial()` more than once, our code will panic!

```
fn main() {
    let serial_1 = unsafe { PERIPHERALS.take_serial() };
    // This panics!
    // let serial_2 = unsafe { PERIPHERALS.take_serial() };
}
```

Although interacting with this structure is `unsafe`, once we have the `SerialPort` it contained, we no longer need to use `unsafe`, or the `PERIPHERALS` structure at all.

This has a small runtime overhead because we must wrap the `SerialPort` structure in an option, and we'll need to call `take_serial()` once, however this small up-front cost allows us to leverage the borrow checker throughout the rest of our program.

Existing library support

Although we created our own `Peripherals` structure above, it is not necessary to do this for your code. the `cortex_m` crate contains a macro called `singleton!()` that will perform this action for you.

```
use cortex_m::singleton;

fn main() {
    // OK if `main` is executed only once
    let x: &'static mut bool =
        singleton!(: bool = false).unwrap();
}
```

cortex_m docs

Additionally, if you use `cortex-m-rtic`, the entire process of defining and obtaining these peripherals are abstracted for you, and you are instead handed a `Peripherals` structure that contains a non-`Option<T>` version of all of the items you define.

```
// cortex-m-rtic v0.5.x
#[rtic::app(device = lm3s6965, peripherals = true)]
const APP: () = {
    #[init]
    fn init(cx: init::Context) {
        static mut X: u32 = 0;

        // Cortex-M peripherals
        let core: cortex_m::Peripherals = cx.core;

        // Device specific peripherals
        let device: lm3s6965::Peripherals = cx.device;
    }
}
```

But why?

But how do these Singletons make a noticeable difference in how our Rust code works?

```
impl SerialPort {
    const SER_PORT_SPEED_REG: *mut u32 = 0x4000_1000 as _;

    fn read_speed(
        &self // <------ This is really, really important
    ) -> u32 {
        unsafe {
            ptr::read_volatile(Self::SER_PORT_SPEED_REG)
        }
    }
}
```

There are two important factors in play here:

- Because we are using a singleton, there is only one way or place to obtain a `SerialPort` structure
- To call the `read_speed()` method, we must have ownership or a reference to a `SerialPort` structure

These two factors put together means that it is only possible to access the hardware if we have appropriately satisfied the borrow checker, meaning that at no point do we have multiple mutable references to the same hardware!

```
fn main() {
    // missing reference to `self`! Won't work.
    // SerialPort::read_speed();

    let serial_1 = unsafe { PERIPHERALS.take_serial() };

    // you can only read what you have access to
    let _ = serial_1.read_speed();
}
```

Treat your hardware like data

Additionally, because some references are mutable, and some are immutable, it becomes possible to see whether a function or method could potentially modify the state of the hardware. For example,

This is allowed to change hardware settings:

```
fn setup_spi_port(
    spi: &mut SpiPort,
    cs_pin: &mut GpioPin
) -> Result<()> {
    // ...
}
```

This isn't:

```
fn read_button(gpio: &GpioPin) -> bool {
    // ...
}
```

This allows us to enforce whether code should or should not make changes to hardware at **compile time**, rather than at runtime. As a note, this generally only works across one application, but for bare metal systems, our software will be compiled into a single application, so this is not usually a restriction.

Static Guarantees

Rust's type system prevents data races at compile time (see `Send` and `Sync` traits). The type system can also be used to check other properties at compile time; reducing the need for runtime checks in some cases.

When applied to embedded programs these *static checks* can be used, for example, to enforce that configuration of I/O interfaces is done properly. For instance, one can design an API where it is only possible to initialize a serial interface by first configuring the pins that will be used by the interface.

One can also statically check that operations, like setting a pin low, can only be performed on correctly configured peripherals. For example, trying to change the output state of a pin configured in floating input mode would raise a compile error.

And, as seen in the previous chapter, the concept of ownership can be applied to peripherals to ensure that only certain parts of a program can modify a peripheral. This *access control* makes software easier to reason about compared to the alternative of treating peripherals as global mutable state.

Typestate Programming

The concept of typestates describes the encoding of information about the current state of an object into the type of that object. Although this can sound a little arcane, if you have used the Builder Pattern in Rust, you have already started using Typestate Programming!

```rust
pub mod foo_module {
    #[derive(Debug)]
    pub struct Foo {
        inner: u32,
    }

    pub struct FooBuilder {
        a: u32,
        b: u32,
    }

    impl FooBuilder {
        pub fn new(starter: u32) -> Self {
            Self {
                a: starter,
                b: starter,
            }
        }

        pub fn double_a(self) -> Self {
            Self {
                a: self.a * 2,
                b: self.b,
            }
        }

        pub fn into_foo(self) -> Foo {
            Foo {
                inner: self.a + self.b,
            }
        }
    }
}

fn main() {
    let x = foo_module::FooBuilder::new(10)
        .double_a()
        .into_foo();

    println!("{:#?}", x);
}
```

In this example, there is no direct way to create a `Foo` object. We must create a `FooBuilder`, and properly initialize it before we can obtain the `Foo` object we want.

This minimal example encodes two states:

- `FooBuilder`, which represents an "unconfigured", or "configuration in process" state
- `Foo`, which represents a "configured", or "ready to use" state.

Strong Types

Because Rust has a Strong Type System, there is no easy way to magically create an instance of `Foo`, or to turn a `FooBuilder` into a `Foo` without calling the `into_foo()` method. Additionally, calling the `into_foo()` method consumes the original `FooBuilder` structure, meaning it can not be reused without the creation of a new instance.

This allows us to represent the states of our system as types, and to include the necessary actions for state transitions into the methods that exchange one type for another. By creating a `FooBuilder`, and exchanging it for a `Foo` object, we have walked through the steps of a basic state machine.

Peripherals as State Machines

The peripherals of a microcontroller can be thought of as set of state machines. For example, the configuration of a simplified GPIO pin could be represented as the following tree of states:

- Disabled
- Enabled
  - Configured as Output
    - Output: High
    - Output: Low
  - Configured as Input
    - Input: High Resistance
    - Input: Pulled Low
    - Input: Pulled High

If the peripheral starts in the `Disabled` mode, to move to the `Input: High Resistance` mode, we must perform the following steps:

1. Disabled
2. Enabled
3. Configured as Input
4. Input: High Resistance

If we wanted to move from `Input: High Resistance` to `Input: Pulled Low`, we must perform the following steps:

1. Input: High Resistance
2. Input: Pulled Low

Similarly, if we want to move a GPIO pin from configured as `Input: Pulled Low` to `Output: High`, we must perform the following steps:

1. Input: Pulled Low
2. Configured as Input
3. Configured as Output
4. Output: High

Hardware Representation

Typically the states listed above are set by writing values to given registers mapped to a GPIO peripheral. Let's define an imaginary GPIO Configuration Register to illustrate this:

Name	Bit Number(s)	Value	Meaning	Notes
enable	0	0	disabled	Disables the GPIO
		1	enabled	Enables the GPIO
direction	1	0	input	Sets the direction to Input
		1	output	Sets the direction to Output
input_mode	2..3	00	hi-z	Sets the input as high resistance
		01	pull-low	Input pin is pulled low
		10	pull-high	Input pin is pulled high
		11	n/a	Invalid state. Do not set
output_mode	4	0	set-low	Output pin is driven low
		1	set-high	Output pin is driven high
input_status	5	x	in-val	0 if input is < 1.5v, 1 if input >= 1.5v

We *could* expose the following structure in Rust to control this GPIO:

```rust
/// GPIO interface
struct GpioConfig {
    /// GPIO Configuration structure generated by svd2rust
    periph: GPIO_CONFIG,
}

impl GpioConfig {
    pub fn set_enable(&mut self, is_enabled: bool) {
        self.periph.modify(|_r, w| {
            w.enable().set_bit(is_enabled)
        });
    }

    pub fn set_direction(&mut self, is_output: bool) {
        self.periph.modify(|_r, w| {
            w.direction().set_bit(is_output)
        });
    }

    pub fn set_input_mode(&mut self, variant: InputMode) {
        self.periph.modify(|_r, w| {
            w.input_mode().variant(variant)
        });
    }

    pub fn set_output_mode(&mut self, is_high: bool) {
        self.periph.modify(|_r, w| {
            w.output_mode.set_bit(is_high)
        });
    }

    pub fn get_input_status(&self) -> bool {
        self.periph.read().input_status().bit_is_set()
    }
}
```

However, this would allow us to modify certain registers that do not make sense. For example, what happens if we set the `output_mode` field when our GPIO is configured as an input?

In general, use of this structure would allow us to reach states not defined by our state machine above: e.g. an output that is pulled low, or an input that is set high. For some hardware, this may not matter. On other hardware, it could cause unexpected or undefined behavior!

Although this interface is convenient to write, it doesn't enforce the design contracts set out by our hardware implementation.

Design Contracts

In our last chapter, we wrote an interface that *didn't* enforce design contracts. Let's take another look at our imaginary GPIO configuration register:

Name	Bit Number(s)	Value	Meaning	Notes
enable	0	0	disabled	Disables the GPIO
		1	enabled	Enables the GPIO
direction	1	0	input	Sets the direction to Input
		1	output	Sets the direction to Output
input_mode	2..3	00	hi-z	Sets the input as high resistance
		01	pull-low	Input pin is pulled low
		10	pull-high	Input pin is pulled high
		11	n/a	Invalid state. Do not set
output_mode	4	0	set-low	Output pin is driven low
		1	set-high	Output pin is driven high
input_status	5	x	in-val	0 if input is < 1.5v, 1 if input >= 1.5v

If we instead checked the state before making use of the underlying hardware, enforcing our design contracts at runtime, we might write code that looks like this instead:

```rust
/// GPIO interface
struct GpioConfig {
    /// GPIO Configuration structure generated by svd2rust
    periph: GPIO_CONFIG,
}

impl GpioConfig {
    pub fn set_enable(&mut self, is_enabled: bool) {
        self.periph.modify(|_r, w| {
            w.enable().set_bit(is_enabled)
        });
    }

    pub fn set_direction(&mut self, is_output: bool) -> Result<(), ()> {
        if self.periph.read().enable().bit_is_clear() {
            // Must be enabled to set direction
            return Err(());
        }

        self.periph.modify(|r, w| {
            w.direction().set_bit(is_output)
        });

        Ok(())
    }

    pub fn set_input_mode(&mut self, variant: InputMode) -> Result<(), ()> {
        if self.periph.read().enable().bit_is_clear() {
            // Must be enabled to set input mode
            return Err(());
        }

        if self.periph.read().direction().bit_is_set() {
            // Direction must be input
            return Err(());
        }

        self.periph.modify(|_r, w| {
            w.input_mode().variant(variant)
        });

        Ok(())
    }

    pub fn set_output_status(&mut self, is_high: bool) -> Result<(), ()> {
        if self.periph.read().enable().bit_is_clear() {
            // Must be enabled to set output status
            return Err(());
        }

        if self.periph.read().direction().bit_is_clear() {
            // Direction must be output
            return Err(());
```

```
        }

        self.periph.modify(|_r, w| {
            w.output_mode.set_bit(is_high)
        });

        Ok(())
    }

    pub fn get_input_status(&self) -> Result<bool, ()> {
        if self.periph.read().enable().bit_is_clear() {
            // Must be enabled to get status
            return Err(());
        }

        if self.periph.read().direction().bit_is_set() {
            // Direction must be input
            return Err(());
        }

        Ok(self.periph.read().input_status().bit_is_set())
    }
}
```

Because we need to enforce the restrictions on the hardware, we end up doing a lot of runtime checking which wastes time and resources, and this code will be much less pleasant for the developer to use.

Type States

But what if instead, we used Rust's type system to enforce the state transition rules? Take this example:

```
/// GPIO interface
struct GpioConfig<ENABLED, DIRECTION, MODE> {
    /// GPIO Configuration structure generated by svd2rust
    periph: GPIO_CONFIG,
    enabled: ENABLED,
    direction: DIRECTION,
    mode: MODE,
}

// Type states for MODE in GpioConfig
struct Disabled;
struct Enabled;
struct Output;
struct Input;
struct PulledLow;
struct PulledHigh;
struct HighZ;
struct DontCare;

/// These functions may be used on any GPIO Pin
impl<EN, DIR, IN_MODE> GpioConfig<EN, DIR, IN_MODE> {
    pub fn into_disabled(self) -> GpioConfig<Disabled, DontCare, DontCare> {
        self.periph.modify(|_r, w| w.enable.disabled());
        GpioConfig {
            periph: self.periph,
            enabled: Disabled,
            direction: DontCare,
            mode: DontCare,
        }
    }

    pub fn into_enabled_input(self) -> GpioConfig<Enabled, Input, HighZ> {
        self.periph.modify(|_r, w| {
            w.enable.enabled()
             .direction.input()
             .input_mode.high_z()
        });
        GpioConfig {
            periph: self.periph,
            enabled: Enabled,
            direction: Input,
            mode: HighZ,
        }
    }

    pub fn into_enabled_output(self) -> GpioConfig<Enabled, Output, DontCare> {
        self.periph.modify(|_r, w| {
            w.enable.enabled()
             .direction.output()
             .input_mode.set_high()
        });
        GpioConfig {
            periph: self.periph,
```

```
            enabled: Enabled,
            direction: Output,
            mode: DontCare,
        }
    }
}

/// This function may be used on an Output Pin
impl GpioConfig<Enabled, Output, DontCare> {
    pub fn set_bit(&mut self, set_high: bool) {
        self.periph.modify(|_r, w| w.output_mode.set_bit(set_high));
    }
}

/// These methods may be used on any enabled input GPIO
impl<IN_MODE> GpioConfig<Enabled, Input, IN_MODE> {
    pub fn bit_is_set(&self) -> bool {
        self.periph.read().input_status.bit_is_set()
    }

    pub fn into_input_high_z(self) -> GpioConfig<Enabled, Input, HighZ> {
        self.periph.modify(|_r, w| w.input_mode().high_z());
        GpioConfig {
            periph: self.periph,
            enabled: Enabled,
            direction: Input,
            mode: HighZ,
        }
    }

    pub fn into_input_pull_down(self) -> GpioConfig<Enabled, Input, PulledLow> {
        self.periph.modify(|_r, w| w.input_mode().pull_low());
        GpioConfig {
            periph: self.periph,
            enabled: Enabled,
            direction: Input,
            mode: PulledLow,
        }
    }

    pub fn into_input_pull_up(self) -> GpioConfig<Enabled, Input, PulledHigh> {
        self.periph.modify(|_r, w| w.input_mode().pull_high());
        GpioConfig {
            periph: self.periph,
            enabled: Enabled,
            direction: Input,
            mode: PulledHigh,
        }
    }
}
```

Now let's see what the code using this would look like:

```
/*
 * Example 1: Unconfigured to High-Z input
 */
let pin: GpioConfig<Disabled, _, _> = get_gpio();

// Can't do this, pin isn't enabled!
// pin.into_input_pull_down();

// Now turn the pin from unconfigured to a high-z input
let input_pin = pin.into_enabled_input();

// Read from the pin
let pin_state = input_pin.bit_is_set();

// Can't do this, input pins don't have this interface!
// input_pin.set_bit(true);

/*
 * Example 2: High-Z input to Pulled Low input
 */
let pulled_low = input_pin.into_input_pull_down();
let pin_state = pulled_low.bit_is_set();

/*
 * Example 3: Pulled Low input to Output, set high
 */
let output_pin = pulled_low.into_enabled_output();
output_pin.set_bit(true);

// Can't do this, output pins don't have this interface!
// output_pin.into_input_pull_down();
```

This is definitely a convenient way to store the state of the pin, but why do it this way? Why is this better than storing the state as an `enum` inside of our `GpioConfig` structure?

Compile Time Functional Safety

Because we are enforcing our design constraints entirely at compile time, this incurs no runtime cost. It is impossible to set an output mode when you have a pin in an input mode. Instead, you must walk through the states by converting it to an output pin, and then setting the output mode. Because of this, there is no runtime penalty due to checking the current state before executing a function.

Also, because these states are enforced by the type system, there is no longer room for errors by consumers of this interface. If they try to perform an illegal state transition, the code will not compile!

Zero Cost Abstractions

Type states are also an excellent example of Zero Cost Abstractions - the ability to move certain behaviors to compile time execution or analysis. These type states contain no actual data, and are instead used as markers. Since they contain no data, they have no actual representation in memory at runtime:

```
use core::mem::size_of;

let _ = size_of::<Enabled>();    // == 0
let _ = size_of::<Input>();      // == 0
let _ = size_of::<PulledHigh>(); // == 0
let _ = size_of::<GpioConfig<Enabled, Input, PulledHigh>>(); // == 0
```

Zero Sized Types

```
struct Enabled;
```

Structures defined like this are called Zero Sized Types, as they contain no actual data. Although these types act "real" at compile time - you can copy them, move them, take references to them, etc., however the optimizer will completely strip them away.

In this snippet of code:

```
pub fn into_input_high_z(self) -> GpioConfig<Enabled, Input, HighZ> {
    self.periph.modify(|_r, w| w.input_mode().high_z());
    GpioConfig {
        periph: self.periph,
        enabled: Enabled,
        direction: Input,
        mode: HighZ,
    }
}
```

The GpioConfig we return never exists at runtime. Calling this function will generally boil down to a single assembly instruction - storing a constant register value to a register location. This means that the type state interface we've developed is a zero cost abstraction - it uses no more CPU, RAM, or code space tracking the state of `GpioConfig`, and renders to the same machine code as a direct register access.

Nesting

In general, these abstractions may be nested as deeply as you would like. As long as all components used are zero sized types, the whole structure will not exist at runtime.

For complex or deeply nested structures, it may be tedious to define all possible combinations of state. In these cases, macros may be used to generate all implementations.

Portability

In embedded environments portability is a very important topic: Every vendor and even each family from a single manufacturer offers different peripherals and capabilities and similarly the ways to interact with the peripherals will vary.

A common way to equalize such differences is via a layer called Hardware Abstraction layer or **HAL**.

> Hardware abstractions are sets of routines in software that emulate some platform-specific details, giving programs direct access to the hardware resources.
>
> They often allow programmers to write device-independent, high performance applications by providing standard operating system (OS) calls to hardware.
>
> *Wikipedia: Hardware Abstraction Layer*

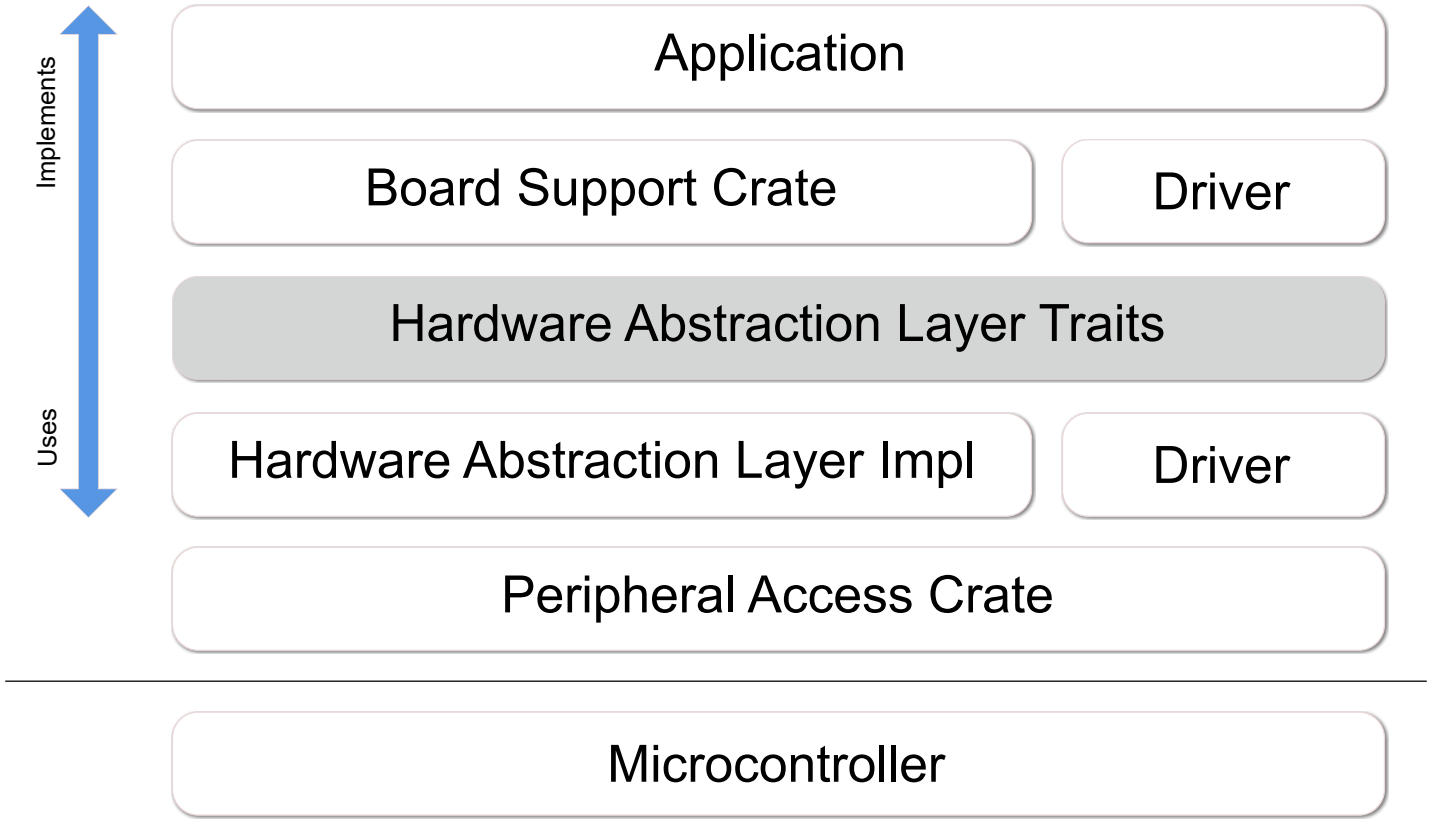
Embedded systems are a bit special in this regard since we typically do not have operating systems and user installable software but firmware images which are compiled as a whole as well as a number of other constraints. So while the traditional approach as defined by Wikipedia could potentially work it is likely not the most productive approach to ensure portability.

How do we do this in Rust? Enter **embedded-hal**...

What is embedded-hal?

In a nutshell it is a set of traits which define implementation contracts between **HAL implementations**, **drivers** and **applications (or firmwares)**. Those contracts include both capabilities (i.e. if a trait is implemented for a certain type, the **HAL implementation** provides a certain capability) and methods (i.e. if you can construct a type implementing a trait it is guaranteed that you have the methods specified in the trait available).

A typical layering might look like this:



Some of the defined traits in **embedded-hal** are:

- GPIO (input and output pins)
- Serial communication
- I2C
- SPI
- Timers/Countdowns
- Analog Digital Conversion

The main reason for having the **embedded-hal** traits and crates implementing and using them is to keep complexity in check. If you consider that an application might have to implement the use of the peripheral in the hardware as well as the application and potentially drivers for additional hardware components, then it should be easy to see that the re-usability is very limited. Expressed mathematically, if **M** is the number of peripheral HAL implementations and **N** the number of drivers then if we were to reinvent the wheel for every application then we would end up with **M*N** implementations while by using the *API* provided by the **embedded-hal** traits will make the implementation complexity approach **M+N**. Of course there're additional benefits to be had, such as less trial-and-error due to a well-defined and ready-to-use APIs.

Users of the embedded-hal

As said above there are three main users of the HAL:

HAL implementation

A HAL implementation provides the interfacing between the hardware and the users of the HAL traits. Typical implementations consist of three parts:

- One or more hardware specific types
- Functions to create and initialize such a type, often providing various configuration options (speed, operation mode, use pins, etc.)
- one or more `trait impl` of **embedded-hal** traits for that type

Such a **HAL implementation** can come in various flavours:

- Via low-level hardware access, e.g. via registers
- Via operating system, e.g. by using the `sysfs` under Linux
- Via adapter, e.g. a mock of types for unit testing
- Via driver for hardware adapters, e.g. I2C multiplexer or GPIO expander

Driver

A driver implements a set of custom functionality for an internal or external component, connected to a peripheral implementing the embedded-hal traits. Typical examples for such drivers include various sensors (temperature, magnetometer, accelerometer, light), display devices (LED arrays, LCD displays) and actuators (motors, transmitters).

A driver has to be initialized with an instance of type that implements a certain `trait` of the embedded-hal which is ensured via trait bound and provides its own type instance with a custom set of methods allowing to interact with the driven device.

Application

The application binds the various parts together and ensures that the desired functionality is achieved. When porting between different systems, this is the part which requires the most adaptation efforts, since the application needs to correctly initialize the real hardware via the HAL implementation and the initialisation of different hardware differs, sometimes drastically so. Also the user choice often plays a big role, since components can be physically connected to different terminals, hardware buses sometimes need external hardware to match the

configuration or there are different trade-offs to be made in the use of internal peripherals (e.g. multiple timers with different capabilities are available or peripherals conflict with others).

Concurrency

Concurrency happens whenever different parts of your program might execute at different times or out of order. In an embedded context, this includes:

- interrupt handlers, which run whenever the associated interrupt happens,
- various forms of multithreading, where your microprocessor regularly swaps between parts of your program,
- and in some systems, multiple-core microprocessors, where each core can be independently running a different part of your program at the same time.

Since many embedded programs need to deal with interrupts, concurrency will usually come up sooner or later, and it's also where many subtle and difficult bugs can occur. Luckily, Rust provides a number of abstractions and safety guarantees to help us write correct code.

No Concurrency

The simplest concurrency for an embedded program is no concurrency: your software consists of a single main loop which just keeps running, and there are no interrupts at all. Sometimes this is perfectly suited to the problem at hand! Typically your loop will read some inputs, perform some processing, and write some outputs.

```
#[entry]
fn main() {
    let peripherals = setup_peripherals();
    loop {
        let inputs = read_inputs(&peripherals);
        let outputs = process(inputs);
        write_outputs(&peripherals, outputs);
    }
}
```

Since there's no concurrency, there's no need to worry about sharing data between parts of your program or synchronising access to peripherals. If you can get away with such a simple approach this can be a great solution.

Global Mutable Data

Unlike non-embedded Rust, we will not usually have the luxury of creating heap allocations and passing references to that data into a newly-created thread. Instead, our interrupt handlers might be called at any time and must know how to access whatever shared memory we are using. At the lowest level, this means we must have *statically allocated* mutable memory, which both the interrupt handler and the main code can refer to.

In Rust, such `static mut` variables are always unsafe to read or write, because without taking special care, you might trigger a race condition, where your access to the variable is interrupted halfway through by an interrupt which also accesses that variable.

For an example of how this behaviour can cause subtle errors in your code, consider an embedded program which counts rising edges of some input signal in each one-second period (a frequency counter):

```
static mut COUNTER: u32 = 0;

#[entry]
fn main() -> ! {
    set_timer_1hz();
    let mut last_state = false;
    loop {
        let state = read_signal_level();
        if state && !last_state {
            // DANGER - Not actually safe! Could cause data races.
            unsafe { COUNTER += 1 };
        }
        last_state = state;
    }
}

#[interrupt]
fn timer() {
    unsafe { COUNTER = 0; }
}
```

Each second, the timer interrupt sets the counter back to 0. Meanwhile, the main loop continually measures the signal, and incremements the counter when it sees a change from low to high. We've had to use `unsafe` to access `COUNTER`, as it's `static mut`, and that means we're promising the compiler we won't cause any undefined behaviour. Can you spot the race condition? The increment on `COUNTER` is *not* guaranteed to be atomic — in fact, on most embedded platforms, it will be split into a load, then the increment, then a store. If the interrupt fired after the load but before the store, the reset back to 0 would be ignored after the interrupt returns — and we would count twice as many transitions for that period.

Critical Sections

So, what can we do about data races? A simple approach is to use *critical sections*, a context where interrupts are disabled. By wrapping the access to `COUNTER` in `main` in a critical section, we can be sure the timer interrupt will not fire until we're finished incrementing `COUNTER`:

```
static mut COUNTER: u32 = 0;

#[entry]
fn main() -> ! {
    set_timer_1hz();
    let mut last_state = false;
    loop {
        let state = read_signal_level();
        if state && !last_state {
            // New critical section ensures synchronised access to COUNTER
            cortex_m::interrupt::free(|_| {
                unsafe { COUNTER += 1 };
            });
        }
        last_state = state;
    }
}

#[interrupt]
fn timer() {
    unsafe { COUNTER = 0; }
}
```

In this example, we use `cortex_m::interrupt::free`, but other platforms will have similar mechanisms for executing code in a critical section. This is also the same as disabling interrupts, running some code, and then re-enabling interrupts.

Note we didn't need to put a critical section inside the timer interrupt, for two reasons:

- Writing 0 to `COUNTER` can't be affected by a race since we don't read it
- It will never be interrupted by the `main` thread anyway

If `COUNTER` was being shared by multiple interrupt handlers that might *preempt* each other, then each one might require a critical section as well.

This solves our immediate problem, but we're still left writing a lot of unsafe code which we need to carefully reason about, and we might be using critical sections needlessly. Since each critical section temporarily pauses interrupt processing, there is an associated cost of some extra code size and higher interrupt latency and jitter (interrupts may take longer to be processed, and the time until they are processed will be more variable). Whether this is a problem depends on your system, but in general, we'd like to avoid it.

It's worth noting that while a critical section guarantees no interrupts will fire, it does not provide an exclusivity guarantee on multi-core systems! The other core could be happily accessing the same memory as your core, even without interrupts. You will need stronger synchronisation primitives if you are using multiple cores.

Atomic Access

On some platforms, special atomic instructions are available, which provide guarantees about read-modify-write operations. Specifically for Cortex-M: `thumbv6` (Cortex-M0, Cortex-M0+) only provide atomic load and store instructions, while `thumbv7` (Cortex-M3 and above) provide full Compare and Swap (CAS) instructions. These CAS instructions give an alternative to the heavy-handed disabling of all interrupts: we can attempt the increment, it will succeed most of the time, but if it was interrupted it will automatically retry the entire increment operation. These atomic operations are safe even across multiple cores.

```
use core::sync::atomic::{AtomicUsize, Ordering};

static COUNTER: AtomicUsize = AtomicUsize::new(0);

#[entry]
fn main() -> ! {
    set_timer_1hz();
    let mut last_state = false;
    loop {
        let state = read_signal_level();
        if state && !last_state {
            // Use `fetch_add` to atomically add 1 to COUNTER
            COUNTER.fetch_add(1, Ordering::Relaxed);
        }
        last_state = state;
    }
}

#[interrupt]
fn timer() {
    // Use `store` to write 0 directly to COUNTER
    COUNTER.store(0, Ordering::Relaxed)
}
```

This time `COUNTER` is a safe `static` variable. Thanks to the `AtomicUsize` type `COUNTER` can be safely modified from both the interrupt handler and the main thread without disabling interrupts. When possible, this is a better solution — but it may not be supported on your platform.

A note on `Ordering`: this affects how the compiler and hardware may reorder instructions, and also has consequences on cache visibility. Assuming that the target is a single core platform `Relaxed` is sufficient and the most efficient choice in this particular case. Stricter ordering will cause the compiler to emit memory barriers around the atomic operations; depending on what you're using atomics for you may or may not need this! The precise details of the atomic model are complicated and best described elsewhere.

For more details on atomics and ordering, see the nomicon.

Abstractions, Send, and Sync

None of the above solutions are especially satisfactory. They require `unsafe` blocks which must be very carefully checked and are not ergonomic. Surely we can do better in Rust!

We can abstract our counter into a safe interface which can be safely used anywhere else in our code. For this example, we'll use the critical-section counter, but you could do something very similar with atomics.

```rust
use core::cell::UnsafeCell;
use cortex_m::interrupt;

// Our counter is just a wrapper around UnsafeCell<u32>, which is the heart
// of interior mutability in Rust. By using interior mutability, we can have
// COUNTER be `static` instead of `static mut`, but still able to mutate
// its counter value.
struct CSCounter(UnsafeCell<u32>);

const CS_COUNTER_INIT: CSCounter = CSCounter(UnsafeCell::new(0));

impl CSCounter {
    pub fn reset(&self, _cs: &interrupt::CriticalSection) {
        // By requiring a CriticalSection be passed in, we know we must
        // be operating inside a CriticalSection, and so can confidently
        // use this unsafe block (required to call UnsafeCell::get).
        unsafe { *self.0.get() = 0 };
    }

    pub fn increment(&self, _cs: &interrupt::CriticalSection) {
        unsafe { *self.0.get() += 1 };
    }
}

// Required to allow static CSCounter. See explanation below.
unsafe impl Sync for CSCounter {}

// COUNTER is no longer `mut` as it uses interior mutability;
// therefore it also no longer requires unsafe blocks to access.
static COUNTER: CSCounter = CS_COUNTER_INIT;

#[entry]
fn main() -> ! {
    set_timer_1hz();
    let mut last_state = false;
    loop {
        let state = read_signal_level();
        if state && !last_state {
            // No unsafe here!
            interrupt::free(|cs| COUNTER.increment(cs));
        }
        last_state = state;
    }
}

#[interrupt]
fn timer() {
    // We do need to enter a critical section here just to obtain a valid
    // cs token, even though we know no other interrupt could pre-empt
    // this one.
    interrupt::free(|cs| COUNTER.reset(cs));

    // We could use unsafe code to generate a fake CriticalSection if we
```

```
    // really wanted to, avoiding the overhead:
    // let cs = unsafe { interrupt::CriticalSection::new() };
}
```

We've moved our `unsafe` code to inside our carefully-planned abstraction, and now our application code does not contain any `unsafe` blocks.

This design requires that the application pass a `CriticalSection` token in: these tokens are only safely generated by `interrupt::free`, so by requiring one be passed in, we ensure we are operating inside a critical section, without having to actually do the lock ourselves. This guarantee is provided statically by the compiler: there won't be any runtime overhead associated with `cs`. If we had multiple counters, they could all be given the same `cs`, without requiring multiple nested critical sections.

This also brings up an important topic for concurrency in Rust: the `Send` and `Sync` traits. To summarise the Rust book, a type is Send when it can safely be moved to another thread, while it is Sync when it can be safely shared between multiple threads. In an embedded context, we consider interrupts to be executing in a separate thread to the application code, so variables accessed by both an interrupt and the main code must be Sync.

For most types in Rust, both of these traits are automatically derived for you by the compiler. However, because `CSCounter` contains an `UnsafeCell`, it is not Sync, and therefore we could not make a `static CSCounter`: `static` variables *must* be Sync, since they can be accessed by multiple threads.

To tell the compiler we have taken care that the `CSCounter` is in fact safe to share between threads, we implement the Sync trait explicitly. As with the previous use of critical sections, this is only safe on single-core platforms: with multiple cores, you would need to go to greater lengths to ensure safety.

Mutexes

We've created a useful abstraction specific to our counter problem, but there are many common abstractions used for concurrency.

One such *synchronisation primitive* is a mutex, short for mutual exclusion. These constructs ensure exclusive access to a variable, such as our counter. A thread can attempt to *lock* (or *acquire*) the mutex, and either succeeds immediately, or blocks waiting for the lock to be acquired, or returns an error that the mutex could not be locked. While that thread holds the lock, it is granted access to the protected data. When the thread is done, it *unlocks* (or *releases*) the mutex, allowing another thread to lock it. In Rust, we would usually implement the unlock using the `Drop` trait to ensure it is always released when the mutex goes out of scope.

Using a mutex with interrupt handlers can be tricky: it is not normally acceptable for the interrupt handler to block, and it would be especially disastrous for it to block waiting for the main thread to release a lock, since we would then *deadlock* (the main thread will never release the lock because execution stays in the interrupt handler). Deadlocking is not considered unsafe: it is possible even in safe Rust.

To avoid this behaviour entirely, we could implement a mutex which requires a critical section to lock, just like our counter example. So long as the critical section must last as long as the lock, we can be sure we have exclusive access to the wrapped variable without even needing to track the lock/unlock state of the mutex.

This is in fact done for us in the `cortex_m` crate! We could have written our counter using it:

```
use core::cell::Cell;
use cortex_m::interrupt::Mutex;

static COUNTER: Mutex<Cell<u32>> = Mutex::new(Cell::new(0));

#[entry]
fn main() -> ! {
    set_timer_1hz();
    let mut last_state = false;
    loop {
        let state = read_signal_level();
        if state && !last_state {
            interrupt::free(|cs|
                COUNTER.borrow(cs).set(COUNTER.borrow(cs).get() + 1));
        }
        last_state = state;
    }
}

#[interrupt]
fn timer() {
    // We still need to enter a critical section here to satisfy the Mutex.
    interrupt::free(|cs| COUNTER.borrow(cs).set(0));
}
```

We're now using `Cell`, which along with its sibling `RefCell` is used to provide safe interior mutability. We've already seen `UnsafeCell` which is the bottom layer of interior mutability in Rust: it allows you to obtain multiple mutable references to its value, but only with unsafe code. A `Cell` is like an `UnsafeCell` but it provides a safe interface: it only permits taking a copy of the current value or replacing it, not taking a reference, and since it is not Sync, it cannot be shared between threads. These constraints mean it's safe to use, but we couldn't use it directly in a `static` variable as a `static` must be Sync.

So why does the example above work? The `Mutex<T>` implements Sync for any `T` which is Send — such as a `Cell`. It can do this safely because it only gives access to its contents during

a critical section. We're therefore able to get a safe counter with no unsafe code at all!

This is great for simple types like the `u32` of our counter, but what about more complex types which are not Copy? An extremely common example in an embedded context is a peripheral struct, which generally is not Copy. For that, we can turn to `RefCell`.

Sharing Peripherals

Device crates generated using `svd2rust` and similar abstractions provide safe access to peripherals by enforcing that only one instance of the peripheral struct can exist at a time. This ensures safety, but makes it difficult to access a peripheral from both the main thread and an interrupt handler.

To safely share peripheral access, we can use the `Mutex` we saw before. We'll also need to use `RefCell`, which uses a runtime check to ensure only one reference to a peripheral is given out at a time. This has more overhead than the plain `Cell`, but since we are giving out references rather than copies, we must be sure only one exists at a time.

Finally, we'll also have to account for somehow moving the peripheral into the shared variable after it has been initialised in the main code. To do this we can use the `Option` type, initialised to `None` and later set to the instance of the peripheral.

```rust
use core::cell::RefCell;
use cortex_m::interrupt::{self, Mutex};
use stm32f4::stm32f405;

static MY_GPIO: Mutex<RefCell<Option<stm32f405::GPIOA>>> =
    Mutex::new(RefCell::new(None));

#[entry]
fn main() -> ! {
    // Obtain the peripheral singletons and configure it.
    // This example is from an svd2rust-generated crate, but
    // most embedded device crates will be similar.
    let dp = stm32f405::Peripherals::take().unwrap();
    let gpioa = &dp.GPIOA;

    // Some sort of configuration function.
    // Assume it sets PA0 to an input and PA1 to an output.
    configure_gpio(gpioa);

    // Store the GPIOA in the mutex, moving it.
    interrupt::free(|cs| MY_GPIO.borrow(cs).replace(Some(dp.GPIOA)));
    // We can no longer use `gpioa` or `dp.GPIOA`, and instead have to
    // access it via the mutex.

    // Be careful to enable the interrupt only after setting MY_GPIO:
    // otherwise the interrupt might fire while it still contains None,
    // and as-written (with `unwrap()`), it would panic.
    set_timer_1hz();
    let mut last_state = false;
    loop {
        // We'll now read state as a digital input, via the mutex
        let state = interrupt::free(|cs| {
            let gpioa = MY_GPIO.borrow(cs).borrow();
            gpioa.as_ref().unwrap().idr.read().idr0().bit_is_set()
        });

        if state && !last_state {
            // Set PA1 high if we've seen a rising edge on PA0.
            interrupt::free(|cs| {
                let gpioa = MY_GPIO.borrow(cs).borrow();
                gpioa.as_ref().unwrap().odr.modify(|_, w| w.odr1().set_bit());
            });
        }
        last_state = state;
    }
}

#[interrupt]
fn timer() {
    // This time in the interrupt we'll just clear PA0.
    interrupt::free(|cs| {
        // We can use `unwrap()` because we know the interrupt wasn't enabled
        // until after MY_GPIO was set; otherwise we should handle the potential
```

```
        // for a None value.
        let gpioa = MY_GPIO.borrow(cs).borrow();
        gpioa.as_ref().unwrap().odr.modify(|_, w| w.odr1().clear_bit());
    });
}
```

That's quite a lot to take in, so let's break down the important lines.

```
static MY_GPIO: Mutex<RefCell<Option<stm32f405::GPIOA>>> =
    Mutex::new(RefCell::new(None));
```

Our shared variable is now a `Mutex` around a `RefCell` which contains an `Option`. The `Mutex` ensures we only have access during a critical section, and therefore makes the variable Sync, even though a plain `RefCell` would not be Sync. The `RefCell` gives us interior mutability with references, which we'll need to use our `GPIOA`. The `Option` lets us initialise this variable to something empty, and only later actually move the variable in. We cannot access the peripheral singleton statically, only at runtime, so this is required.

```
interrupt::free(|cs| MY_GPIO.borrow(cs).replace(Some(dp.GPIOA)));
```

Inside a critical section we can call `borrow()` on the mutex, which gives us a reference to the `RefCell`. We then call `replace()` to move our new value into the `RefCell`.

```
interrupt::free(|cs| {
    let gpioa = MY_GPIO.borrow(cs).borrow();
    gpioa.as_ref().unwrap().odr.modify(|_, w| w.odr1().set_bit());
});
```

Finally, we use `MY_GPIO` in a safe and concurrent fashion. The critical section prevents the interrupt firing as usual, and lets us borrow the mutex. The `RefCell` then gives us an `&Option<GPIOA>`, and tracks how long it remains borrowed - once that reference goes out of scope, the `RefCell` will be updated to indicate it is no longer borrowed.

Since we can't move the `GPIOA` out of the `&Option`, we need to convert it to an `&Option<&GPIOA>` with `as_ref()`, which we can finally `unwrap()` to obtain the `&GPIOA` which lets us modify the peripheral.

If we need a mutable reference to a shared resource, then `borrow_mut` and `deref_mut` should be used instead. The following code shows an example using the TIM2 timer.

```rust
use core::cell::RefCell;
use core::ops::DerefMut;
use cortex_m::interrupt::{self, Mutex};
use cortex_m::asm::wfi;
use stm32f4::stm32f405;

static G_TIM: Mutex<RefCell<Option<Timer<stm32::TIM2>>>> =
    Mutex::new(RefCell::new(None));

#[entry]
fn main() -> ! {
    let mut cp = cm::Peripherals::take().unwrap();
    let dp = stm32f405::Peripherals::take().unwrap();

    // Some sort of timer configuration function.
    // Assume it configures the TIM2 timer, its NVIC interrupt,
    // and finally starts the timer.
    let tim = configure_timer_interrupt(&mut cp, dp);

    interrupt::free(|cs| {
        G_TIM.borrow(cs).replace(Some(tim));
    });

    loop {
        wfi();
    }
}

#[interrupt]
fn timer() {
    interrupt::free(|cs| {
        if let Some(ref mut tim)) =  G_TIM.borrow(cs).borrow_mut().deref_mut() {
            tim.start(1.hz());
        }
    });
}
```

Whew! This is safe, but it is also a little unwieldy. Is there anything else we can do?

RTIC

One alternative is the RTIC framework, short for Real Time Interrupt-driven Concurrency. It enforces static priorities and tracks accesses to `static mut` variables ("resources") to statically ensure that shared resources are always accessed safely, without requiring the overhead of always entering critical sections and using reference counting (as in `RefCell`). This has a number of advantages such as guaranteeing no deadlocks and giving extremely low time and memory overhead.

The framework also includes other features like message passing, which reduces the need for explicit shared state, and the ability to schedule tasks to run at a given time, which can be used to implement periodic tasks. Check out the documentation for more information!

Real Time Operating Systems

Another common model for embedded concurrency is the real-time operating system (RTOS). While currently less well explored in Rust, they are widely used in traditional embedded development. Open source examples include FreeRTOS and ChibiOS. These RTOSs provide support for running multiple application threads which the CPU swaps between, either when the threads yield control (called cooperative multitasking) or based on a regular timer or interrupts (preemptive multitasking). The RTOS typically provide mutexes and other synchronisation primitives, and often interoperate with hardware features such as DMA engines.

At the time of writing, there are not many Rust RTOS examples to point to, but it's an interesting area so watch this space!

Multiple Cores

It is becoming more common to have two or more cores in embedded processors, which adds an extra layer of complexity to concurrency. All the examples using a critical section (including the `cortex_m::interrupt::Mutex`) assume the only other execution thread is the interrupt thread, but on a multi-core system that's no longer true. Instead, we'll need synchronisation primitives designed for multiple cores (also called SMP, for symmetric multi-processing).

These typically use the atomic instructions we saw earlier, since the processing system will ensure that atomicity is maintained over all cores.

Covering these topics in detail is currently beyond the scope of this book, but the general patterns are the same as for the single-core case.

Collections

Eventually you'll want to use dynamic data structures (AKA collections) in your program. `std` provides a set of common collections: `Vec`, `String`, `HashMap`, etc. All the collections implemented in `std` use a global dynamic memory allocator (AKA the heap).

As `core` is, by definition, free of memory allocations these implementations are not available there, but they can be found in the `alloc` crate that's shipped with the compiler.

If you need collections, a heap allocated implementation is not your only option. You can also use *fixed capacity* collections; one such implementation can be found in the `heapless` crate.

In this section, we'll explore and compare these two implementations.

Using `alloc`

The `alloc` crate is shipped with the standard Rust distribution. To import the crate you can directly `use` it *without* declaring it as a dependency in your `Cargo.toml` file.

```
#![feature(alloc)]

extern crate alloc;

use alloc::vec::Vec;
```

To be able to use any collection you'll first need use the `global_allocator` attribute to declare the global allocator your program will use. It's required that the allocator you select implements the `GlobalAlloc` trait.

For completeness and to keep this section as self-contained as possible we'll implement a simple bump pointer allocator and use that as the global allocator. However, we *strongly* suggest you use a battle tested allocator from crates.io in your program instead of this allocator.

```rust
// Bump pointer allocator implementation

use core::alloc::{GlobalAlloc, Layout};
use core::cell::UnsafeCell;
use core::ptr;

use cortex_m::interrupt;

// Bump pointer allocator for *single* core systems
struct BumpPointerAlloc {
    head: UnsafeCell<usize>,
    end: usize,
}

unsafe impl Sync for BumpPointerAlloc {}

unsafe impl GlobalAlloc for BumpPointerAlloc {
    unsafe fn alloc(&self, layout: Layout) -> *mut u8 {
        // `interrupt::free` is a critical section that makes our allocator safe
        // to use from within interrupts
        interrupt::free(|_| {
            let head = self.head.get();
            let size = layout.size();
            let align = layout.align();
            let align_mask = !(align - 1);

            // move start up to the next alignment boundary
            let start = (*head + align - 1) & align_mask;

            if start + size > self.end {
                // a null pointer signal an Out Of Memory condition
                ptr::null_mut()
            } else {
                *head = start + size;
                start as *mut u8
            }
        })
    }

    unsafe fn dealloc(&self, _: *mut u8, _: Layout) {
        // this allocator never deallocates memory
    }
}

// Declaration of the global memory allocator
// NOTE the user must ensure that the memory region `[0x2000_0100, 0x2000_0200]`
// is not used by other parts of the program
#[global_allocator]
static HEAP: BumpPointerAlloc = BumpPointerAlloc {
    head: UnsafeCell::new(0x2000_0100),
    end: 0x2000_0200,
};
```

Apart from selecting a global allocator the user will also have to define how Out Of Memory (OOM) errors are handled using the *unstable* `alloc_error_handler` attribute.

```
#![feature(alloc_error_handler)]

use cortex_m::asm;

#[alloc_error_handler]
fn on_oom(_layout: Layout) -> ! {
    asm::bkpt();

    loop {}
}
```

Once all that is in place, the user can finally use the collections in `alloc`.

```
#[entry]
fn main() -> ! {
    let mut xs = Vec::new();

    xs.push(42);
    assert!(xs.pop(), Some(42));

    loop {
        // ..
    }
}
```

If you have used the collections in the `std` crate then these will be familiar as they are exact same implementation.

Using `heapless`

`heapless` requires no setup as its collections don't depend on a global memory allocator. Just `use` its collections and proceed to instantiate them:

```rust
// heapless version: v0.4.x
use heapless::Vec;
use heapless::consts::*;

#[entry]
fn main() -> ! {
    let mut xs: Vec<_, U8> = Vec::new();

    xs.push(42).unwrap();
    assert_eq!(xs.pop(), Some(42));
    loop {}
}
```

You'll note two differences between these collections and the ones in `alloc`.

First, you have to declare upfront the capacity of the collection. `heapless` collections never reallocate and have fixed capacities; this capacity is part of the type signature of the collection. In this case we have declared that `xs` has a capacity of 8 elements that is the vector can, at most, hold 8 elements. This is indicated by the `U8` (see `typenum`) in the type signature.

Second, the `push` method, and many other methods, return a `Result`. Since the `heapless` collections have fixed capacity all operations that insert elements into the collection can potentially fail. The API reflects this problem by returning a `Result` indicating whether the operation succeeded or not. In contrast, `alloc` collections will reallocate themselves on the heap to increase their capacity.

As of version v0.4.x all `heapless` collections store all their elements inline. This means that an operation like `let x = heapless::Vec::new();` will allocate the collection on the stack, but it's also possible to allocate the collection on a `static` variable, or even on the heap (`Box<Vec<_, _>>`).

Trade-offs

Keep these in mind when choosing between heap allocated, relocatable collections and fixed capacity collections.

Out Of Memory and error handling

With heap allocations Out Of Memory is always a possibility and can occur in any place where a collection may need to grow: for example, all `alloc::Vec.push` invocations can potentially generate an OOM condition. Thus some operations can *implicitly* fail. Some `alloc` collections

expose `try_reserve` methods that let you check for potential OOM conditions when growing the collection but you need be proactive about using them.

If you exclusively use `heapless` collections and you don't use a memory allocator for anything else then an OOM condition is impossible. Instead, you'll have to deal with collections running out of capacity on a case by case basis. That is you'll have deal with *all* the `Result`s returned by methods like `Vec.push`.

OOM failures can be harder to debug than say `unwrap`-ing on all `Result`s returned by `heapless::Vec.push` because the observed location of failure may *not* match with the location of the cause of the problem. For example, even `vec.reserve(1)` can trigger an OOM if the allocator is nearly exhausted because some other collection was leaking memory (memory leaks are possible in safe Rust).

Memory usage

Reasoning about memory usage of heap allocated collections is hard because the capacity of long lived collections can change at runtime. Some operations may implicitly reallocate the collection increasing its memory usage, and some collections expose methods like `shrink_to_fit` that can potentially reduce the memory used by the collection -- ultimately, it's up to the allocator to decide whether to actually shrink the memory allocation or not. Additionally, the allocator may have to deal with memory fragmentation which can increase the *apparent* memory usage.

On the other hand if you exclusively use fixed capacity collections, store most of them in `static` variables and set a maximum size for the call stack then the linker will detect if you try to use more memory than what's physically available.

Furthermore, fixed capacity collections allocated on the stack will be reported by `-Z emit-stack-sizes` flag which means that tools that analyze stack usage (like `stack-sizes`) will include them in their analysis.

However, fixed capacity collections can *not* be shrunk which can result in lower load factors (the ratio between the size of the collection and its capacity) than what relocatable collections can achieve.

Worst Case Execution Time (WCET)

If you are building time sensitive applications or hard real time applications then you care, maybe a lot, about the worst case execution time of the different parts of your program.

The `alloc` collections can reallocate so the WCET of operations that may grow the collection will also include the time it takes to reallocate the collection, which itself depends on the *runtime* capacity of the collection. This makes it hard to determine the WCET of, for example, the `alloc::Vec.push` operation as it depends on both the allocator being used and its runtime capacity.

On the other hand fixed capacity collections never reallocate so all operations have a predictable execution time. For example, `heapless::Vec.push` executes in constant time.

Ease of use

`alloc` requires setting up a global allocator whereas `heapless` does not. However, `heapless` requires you to pick the capacity of each collection that you instantiate.

The `alloc` API will be familiar to virtually every Rust developer. The `heapless` API tries to closely mimic the `alloc` API but it will never be exactly the same due to its explicit error handling -- some developers may feel the explicit error handling is excessive or too cumbersome.

Design Patterns

This chapter aims to collect various useful design patterns for embedded Rust.

HAL Design Patterns

This is a set of common and recommended patterns for writing hardware abstraction layers (HALs) for microcontrollers in Rust. These patterns are intended to be used in addition to the existing Rust API Guidelines when writing HALs for microcontrollers.

Checklist

- Naming
- Interoperability
- Predictability
- GPIO

HAL Design Patterns Checklist

- **Naming** *(crate aligns with Rust naming conventions)*
  - ■ The crate is named appropriately (C-CRATE-NAME)
- **Interoperability** *(crate interacts nicely with other library functionality)*
  - ■ Wrapper types provide a destructor method (C-FREE)
  - ■ HALs reexport their register access crate (C-REEXPORT-PAC)
  - ■ Types implement the `embedded-hal` traits (C-HAL-TRAITS)
- **Predictability** *(crate enables legible code that acts how it looks)*
  - ■ Constructors are used instead of extension traits (C-CTOR)
- **GPIO Interfaces** *(GPIO Interfaces follow a common pattern)*
  - ■ Pin types are zero-sized by default (C-ZST-PIN)
  - ■ Pin types provide methods to erase pin and port (C-ERASED-PIN)
  - ■ Pin state should be encoded as type parameters (C-PIN-STATE)

Naming

The crate is named appropriately (C-CRATE-NAME)

HAL crates should be named after the chip or family of chips they aim to support. Their name should end with `-hal` to distinguish them from register access crates. The name should not contain underscores (use dashes instead).

Interoperability

Wrapper types provide a destructor method (C-FREE)

Any non- `Copy` wrapper type provided by the HAL should provide a `free` method that consumes the wrapper and returns back the raw peripheral (and possibly other objects) it was created from.

The method should shut down and reset the peripheral if necessary. Calling `new` with the raw peripheral returned by `free` should not fail due to an unexpected state of the peripheral.

If the HAL type requires other non- `Copy` objects to be constructed (for example I/O pins), any such object should be released and returned by `free` as well. `free` should return a tuple in that case.

For example:

```
pub struct Timer(TIMER0);

impl Timer {
    pub fn new(periph: TIMER0) -> Self {
        Self(periph)
    }

    pub fn free(self) -> TIMER0 {
        self.0
    }
}
```

HALs reexport their register access crate (C-REEXPORT-PAC)

HALs can be written on top of svd2rust-generated PACs, or on top of other crates that provide raw register access. HALs should always reexport the register access crate they are based on in their crate root.

A PAC should be reexported under the name `pac`, regardless of the actual name of the crate, as the name of the HAL should already make it clear what PAC is being accessed.

Types implement the `embedded-hal` traits (C-HAL-TRAITS)

Types provided by the HAL should implement all applicable traits provided by the `embedded-hal` crate.

Multiple traits may be implemented for the same type.

Predictability

Constructors are used instead of extension traits (C-CTOR)

All peripherals to which the HAL adds functionality should be wrapped in a new type, even if no additional fields are required for that functionality.

Extension traits implemented for the raw peripheral should be avoided.

Methods are decorated with `#[inline]` where appropriate (C-INLINE)

The Rust compiler does not by default perform full inlining across crate boundaries. As embedded applications are sensitive to unexpected code size increases, `#[inline]` should be used to guide the compiler as follows:

- All "small" functions should be marked `#[inline]`. What qualifies as "small" is subjective, but generally all functions that are expected to compile down to single-digit instruction sequences qualify as small.
- Functions that are very likely to take constant values as parameters should be marked as `#[inline]`. This enables the compiler to compute even complicated initialization logic at compile time, provided the function inputs are known.

Recommendations for GPIO Interfaces

Pin types are zero-sized by default (C-ZST-PIN)

GPIO Interfaces exposed by the HAL should provide dedicated zero-sized types for each pin on every interface or port, resulting in a zero-cost GPIO abstraction when all pin assignments are statically known.

Each GPIO Interface or Port should implement a `split` method returning a struct with every pin.

Example:

```rust
pub struct PA0;
pub struct PA1;
// ...

pub struct PortA;

impl PortA {
    pub fn split(self) -> PortAPins {
        PortAPins {
            pa0: PA0,
            pa1: PA1,
            // ...
        }
    }
}

pub struct PortAPins {
    pub pa0: PA0,
    pub pa1: PA1,
    // ...
}
```

Pin types provide methods to erase pin and port (C-ERASED-PIN)

Pins should provide type erasure methods that move their properties from compile time to runtime, and allow more flexibility in applications.

Example:

```
/// Port A, pin 0.
pub struct PA0;

impl PA0 {
    pub fn erase_pin(self) -> PA {
        PA { pin: 0 }
    }
}

/// A pin on port A.
pub struct PA {
    /// The pin number.
    pin: u8,
}

impl PA {
    pub fn erase_port(self) -> Pin {
        Pin {
            port: Port::A,
            pin: self.pin,
        }
    }
}

pub struct Pin {
    port: Port,
    pin: u8,
    // (these fields can be packed to reduce the memory footprint)
}

enum Port {
    A,
    B,
    C,
    D,
}
```

Pin state should be encoded as type parameters (C-PIN-STATE)

Pins may be configured as input or output with different characteristics depending on the chip or family. This state should be encoded in the type system to prevent use of pins in incorrect states.

Additional, chip-specific state (eg. drive strength) may also be encoded in this way, using additional type parameters.

Methods for changing the pin state should be provided as `into_input` and `into_output` methods.

Additionally, `with_{input,output}_state` methods should be provided that temporarily reconfigure a pin in a different state without moving it.

The following methods should be provided for every pin type (that is, both erased and non-erased pin types should provide the same API):

- `pub fn into_input<N: InputState>(self, input: N) -> Pin<N>`
- `pub fn into_output<N: OutputState>(self, output: N) -> Pin<N>`

- ```
  pub fn with_input_state<N: InputState, R>(
      &mut self,
      input: N,
      f: impl FnOnce(&mut PA1<N>) -> R,
  ) -> R
  ```

- ```
  pub fn with_output_state<N: OutputState, R>(
      &mut self,
      output: N,
      f: impl FnOnce(&mut PA1<N>) -> R,
  ) -> R
  ```

Pin state should be bounded by sealed traits. Users of the HAL should have no need to add their own state. The traits can provide HAL-specific methods required to implement the pin state API.

Example:

```rust
mod sealed {
    pub trait Sealed {}
}

pub trait PinState: sealed::Sealed {}
pub trait OutputState: sealed::Sealed {}
pub trait InputState: sealed::Sealed {
    // ...
}

pub struct Output<S: OutputState> {
    _p: PhantomData<S>,
}

impl<S: OutputState> PinState for Output<S> {}
impl<S: OutputState> sealed::Sealed for Output<S> {}

pub struct PushPull;
pub struct OpenDrain;

impl OutputState for PushPull {}
impl OutputState for OpenDrain {}
impl sealed::Sealed for PushPull {}
impl sealed::Sealed for OpenDrain {}

pub struct Input<S: InputState> {
    _p: PhantomData<S>,
}

impl<S: InputState> PinState for Input<S> {}
impl<S: InputState> sealed::Sealed for Input<S> {}

pub struct Floating;
pub struct PullUp;
pub struct PullDown;

impl InputState for Floating {}
impl InputState for PullUp {}
impl InputState for PullDown {}
impl sealed::Sealed for Floating {}
impl sealed::Sealed for PullUp {}
impl sealed::Sealed for PullDown {}

pub struct PA1<S: PinState> {
    _p: PhantomData<S>,
}

impl<S: PinState> PA1<S> {
    pub fn into_input<N: InputState>(self, input: N) -> PA1<Input<N>> {
        todo!()
    }

    pub fn into_output<N: OutputState>(self, output: N) -> PA1<Output<N>> {
```

```
            todo!()
        }

        pub fn with_input_state<N: InputState, R>(
            &mut self,
            input: N,
            f: impl FnOnce(&mut PA1<N>) -> R,
        ) -> R {
            todo!()
        }

        pub fn with_output_state<N: OutputState, R>(
            &mut self,
            output: N,
            f: impl FnOnce(&mut PA1<N>) -> R,
        ) -> R {
            todo!()
        }
    }

    // Same for `PA` and `Pin`, and other pin types.
```

Tips for embedded C developers

This chapter collects a variety of tips that might be useful to experienced embedded C developers looking to start writing Rust. It will especially highlight how things you might already be used to in C are different in Rust.

Preprocessor

In embedded C it is very common to use the preprocessor for a variety of purposes, such as:

- Compile-time selection of code blocks with `#ifdef`
- Compile-time array sizes and computations
- Macros to simplify common patterns (to avoid function call overhead)

In Rust there is no preprocessor, and so many of these use cases are addressed differently. In the rest of this section we cover various alternatives to using the preprocessor.

Compile-Time Code Selection

The closest match to `#ifdef ... #endif` in Rust are Cargo features. These are a little more formal than the C preprocessor: all possible features are explicitly listed per crate, and can only be either on or off. Features are turned on when you list a crate as a dependency, and are additive: if any crate in your dependency tree enables a feature for another crate, that feature will be enabled for all users of that crate.

For example, you might have a crate which provides a library of signal processing primitives. Each one might take some extra time to compile or declare some large table of constants which you'd like to avoid. You could declare a Cargo feature for each component in your `Cargo.toml`:

```
[features]
FIR = []
IIR = []
```

Then, in your code, use `#[cfg(feature="FIR")]` to control what is included.

```
/// In your top-level lib.rs

#[cfg(feature="FIR")]
pub mod fir;

#[cfg(feature="IIR")]
pub mod iir;
```

You can similarly include code blocks only if a feature is *not* enabled, or if any combination of features are or are not enabled.

Additionally, Rust provides a number of automatically-set conditions you can use, such as `target_arch` to select different code based on architecture. For full details of the conditional compilation support, refer to the conditional compilation chapter of the Rust reference.

The conditional compilation will only apply to the next statement or block. If a block can not be used in the current scope then the `cfg` attribute will need to be used multiple times. It's worth noting that most of the time it is better to simply include all the code and allow the compiler to remove dead code when optimising: it's simpler for you and your users, and in general the compiler will do a good job of removing unused code.

Compile-Time Sizes and Computation

Rust supports `const fn`, functions which are guaranteed to be evaluable at compile-time and can therefore be used where constants are required, such as in the size of arrays. This can be used alongside features mentioned above, for example:

```
const fn array_size() -> usize {
    #[cfg(feature="use_more_ram")]
    { 1024 }
    #[cfg(not(feature="use_more_ram"))]
    { 128 }
}

static BUF: [u32; array_size()] = [0u32; array_size()];
```

These are new to stable Rust as of 1.31, so documentation is still sparse. The functionality available to `const fn` is also very limited at the time of writing; in future Rust releases it is expected to expand on what is permitted in a `const fn`.

Macros

Rust provides an extremely powerful macro system. While the C preprocessor operates almost directly on the text of your source code, the Rust macro system operates at a higher level. There are two varieties of Rust macro: *macros by example* and *procedural macros*. The former are simpler and most common; they look like function calls and can expand to a complete expression, statement, item, or pattern. Procedural macros are more complex but permit extremely powerful additions to the Rust language: they can transform arbitrary Rust syntax into new Rust syntax.

In general, where you might have used a C preprocessor macro, you probably want to see if a macro-by-example can do the job instead. They can be defined in your crate and easily used by your own crate or exported for other users. Be aware that since they must expand to complete expressions, statements, items, or patterns, some use cases of C preprocessor macros will not work, for example a macro that expands to part of a variable name or an incomplete set of items in a list.

As with Cargo features, it is worth considering if you even need the macro. In many cases a regular function is easier to understand and will be inlined to the same code as a macro. The `#[inline]` and `#[inline(always)]` attributes give you further control over this process, although care should be taken here as well — the compiler will automatically inline functions from the same crate where appropriate, so forcing it to do so inappropriately might actually lead to decreased performance.

Explaining the entire Rust macro system is out of scope for this tips page, so you are encouraged to consult the Rust documentation for full details.

Build System

Most Rust crates are built using Cargo (although it is not required). This takes care of many difficult problems with traditional build systems. However, you may wish to customise the build process. Cargo provides `build.rs` scripts for this purpose. They are Rust scripts which can interact with the Cargo build system as required.

Common use cases for build scripts include:

- provide build-time information, for example statically embedding the build date or Git commit hash into your executable
- generate linker scripts at build time depending on selected features or other logic
- change the Cargo build configuration
- add extra static libraries to link against

At present there is no support for post-build scripts, which you might traditionally have used for tasks like automatic generation of binaries from the build objects or printing build information.

Cross-Compiling

Using Cargo for your build system also simplifies cross-compiling. In most cases it suffices to tell Cargo `--target thumbv6m-none-eabi` and find a suitable executable in `target/thumbv6m-none-eabi/debug/myapp`.

For platforms not natively supported by Rust, you will need to build `libcore` for that target yourself. On such platforms, Xargo can be used as a stand-in for Cargo which automatically builds `libcore` for you.

Iterators vs Array Access

In C you are probably used to accessing arrays directly by their index:

```
int16_t arr[16];
int i;
for(i=0; i<sizeof(arr)/sizeof(arr[0]); i++) {
    process(arr[i]);
}
```

In Rust this is an anti-pattern: indexed access can be slower (as it needs to be bounds checked) and may prevent various compiler optimisations. This is an important distinction and worth repeating: Rust will check for out-of-bounds access on manual array indexing to guarantee memory safety, while C will happily index outside the array.

Instead, use iterators:

```
let arr = [0u16; 16];
for element in arr.iter() {
    process(*element);
}
```

Iterators provide a powerful array of functionality you would have to implement manually in C, such as chaining, zipping, enumerating, finding the min or max, summing, and more. Iterator methods can also be chained, giving very readable data processing code.

See the Iterators in the Book and Iterator documentation for more details.

References vs Pointers

In Rust, pointers (called *raw pointers*) exist but are only used in specific circumstances, as dereferencing them is always considered `unsafe` -- Rust cannot provide its usual guarantees about what might be behind the pointer.

In most cases, we instead use *references*, indicated by the `&` symbol, or *mutable references*, indicated by `&mut`. References behave similarly to pointers, in that they can be dereferenced to access the underlying values, but they are a key part of Rust's ownership system: Rust will strictly enforce that you may only have one mutable reference *or* multiple non-mutable references to the same value at any given time.

In practice this means you have to be more careful about whether you need mutable access to data: where in C the default is mutable and you must be explicit about `const`, in Rust the opposite is true.

One situation where you might still use raw pointers is interacting directly with hardware (for example, writing a pointer to a buffer into a DMA peripheral register), and they are also used under the hood for all peripheral access crates to allow you to read and write memory-mapped registers.

Volatile Access

In C, individual variables may be marked `volatile`, indicating to the compiler that the value in the variable may change between accesses. Volatile variables are commonly used in an embedded context for memory-mapped registers.

In Rust, instead of marking a variable as `volatile`, we use specific methods to perform volatile access: `core::ptr::read_volatile` and `core::ptr::write_volatile`. These methods take a `*const T` or a `*mut T` (*raw pointers*, as discussed above) and perform a volatile read or write.

For example, in C you might write:

```
volatile bool signalled = false;

void ISR() {
    // Signal that the interrupt has occurred
    signalled = true;
}

void driver() {
    while(true) {
        // Sleep until signalled
        while(!signalled) { WFI(); }
        // Reset signalled indicator
        signalled = false;
        // Perform some task that was waiting for the interrupt
        run_task();
    }
}
```

The equivalent in Rust would use volatile methods on each access:

```
static mut SIGNALLED: bool = false;

#[interrupt]
fn ISR() {
    // Signal that the interrupt has occurred
    // (In real code, you should consider a higher level primitive,
    //  such as an atomic type).
    unsafe { core::ptr::write_volatile(&mut SIGNALLED, true) };
}

fn driver() {
    loop {
        // Sleep until signalled
        while unsafe { !core::ptr::read_volatile(&SIGNALLED) } {}
        // Reset signalled indicator
        unsafe { core::ptr::write_volatile(&mut SIGNALLED, false) };
        // Perform some task that was waiting for the interrupt
        run_task();
    }
}
```

A few things are worth noting in the code sample:

- We can pass `&mut SIGNALLED` into the function requiring `*mut T`, since `&mut T` automatically converts to a `*mut T` (and the same for `*const T`)
- We need `unsafe` blocks for the `read_volatile` / `write_volatile` methods, since they are `unsafe` functions. It is the programmer's responsibility to ensure safe use: see the methods' documentation for further details.

It is rare to require these functions directly in your code, as they will usually be taken care of for you by higher-level libraries. For memory mapped peripherals, the peripheral access crates will implement volatile access automatically, while for concurrency primitives there are better abstractions available (see the Concurrency chapter).

Packed and Aligned Types

In embedded C it is common to tell the compiler a variable must have a certain alignment or a struct must be packed rather than aligned, usually to meet specific hardware or protocol requirements.

In Rust this is controlled by the `repr` attribute on a struct or union. The default representation provides no guarantees of layout, so should not be used for code that interoperates with hardware or C. The compiler may re-order struct members or insert padding and the behaviour may change with future versions of Rust.

```
struct Foo {
    x: u16,
    y: u8,
    z: u16,
}

fn main() {
    let v = Foo { x: 0, y: 0, z: 0 };
    println!("{:p} {:p} {:p}", &v.x, &v.y, &v.z);
}

// 0x7ffecb3511d0 0x7ffecb3511d4 0x7ffecb3511d2
// Note ordering has been changed to x, z, y to improve packing.
```

To ensure layouts that are interoperable with C, use `repr(C)`:

```rust
#[repr(C)]
struct Foo {
    x: u16,
    y: u8,
    z: u16,
}

fn main() {
    let v = Foo { x: 0, y: 0, z: 0 };
    println!("{:p} {:p} {:p}", &v.x, &v.y, &v.z);
}

// 0x7fffd0d84c60 0x7fffd0d84c62 0x7fffd0d84c64
// Ordering is preserved and the layout will not change over time.
// `z` is two-byte aligned so a byte of padding exists between `y` and `z`.
```

To ensure a packed representation, use `repr(packed)`:

```rust
#[repr(packed)]
struct Foo {
    x: u16,
    y: u8,
    z: u16,
}

fn main() {
    let v = Foo { x: 0, y: 0, z: 0 };
    // References must always be aligned, so to check the addresses of the
    // struct's fields, we use `std::ptr::addr_of!()` to get a raw pointer
    // instead of just printing `&v.x`.
    let px = std::ptr::addr_of!(v.x);
    let py = std::ptr::addr_of!(v.y);
    let pz = std::ptr::addr_of!(v.z);
    println!("{:p} {:p} {:p}", px, py, pz);
}

// 0x7ffd33598490 0x7ffd33598492 0x7ffd33598493
// No padding has been inserted between `y` and `z`, so now `z` is unaligned.
```

Note that using `repr(packed)` also sets the alignment of the type to `1`.

Finally, to specify a specific alignment, use `repr(align(n))`, where `n` is the number of bytes to align to (and must be a power of two):

```rust
#[repr(C)]
#[repr(align(4096))]
struct Foo {
    x: u16,
    y: u8,
    z: u16,
}

fn main() {
    let v = Foo { x: 0, y: 0, z: 0 };
    let u = Foo { x: 0, y: 0, z: 0 };
    println!("{:p} {:p} {:p}", &v.x, &v.y, &v.z);
    println!("{:p} {:p} {:p}", &u.x, &u.y, &u.z);
}

// 0x7ffec909a000 0x7ffec909a002 0x7ffec909a004
// 0x7ffec909b000 0x7ffec909b002 0x7ffec909b004
// The two instances `u` and `v` have been placed on 4096-byte alignments,
// evidenced by the `000` at the end of their addresses.
```

Note we can combine `repr(C)` with `repr(align(n))` to obtain an aligned and C-compatible layout. It is not permissible to combine `repr(align(n))` with `repr(packed)`, since `repr(packed)` sets the alignment to `1`. It is also not permissible for a `repr(packed)` type to contain a `repr(align(n))` type.

For further details on type layouts, refer to the type layout chapter of the Rust Reference.

Other Resources

- In this book:
  - A little C with your Rust
  - A little Rust with your C
- The Rust Embedded FAQs
- Rust Pointers for C Programmers
- I used to use pointers - now what?

Interoperability

Interoperability between Rust and C code is always dependent on transforming data between the two languages. For this purpose, there is a dedicated module in the `stdlib` called `std::ffi`.

`std::ffi` provides type definitions for C primitive types, such as `char`, `int`, and `long`. It also provides some utility for converting more complex types such as strings, mapping both `&str` and `String` to C types that are easier and safer to handle.

As of Rust 1.30, functionalities of `std::ffi` are available in either `core::ffi` or `alloc::ffi` depending on whether or not memory allocation is involved. The `cty` crate and the `cstr_core` crate also offer similar functionalities.

Rust type	Intermediate	C type
`String`	`CString`	`char *`
`&str`	`CStr`	`const char *`
`()`	`c_void`	`void`
`u32` or `u64`	`c_uint`	`unsigned int`
etc	...	...

A value of a C primitive type can be used as one of the corresponding Rust type and vice versa, since the former is simply a type alias of the latter. For example, the following code compiles on platforms where `unsigned int` is 32-bit long.

```
fn foo(num: u32) {
    let c_num: c_uint = num;
    let r_num: u32 = c_num;
}
```

Interoperability with other build systems

A common requirement for including Rust in your embedded project is combining Cargo with your existing build system, such as make or cmake.

We are collecting examples and use cases for this on our issue tracker in issue #61.

Interoperability with RTOSs

Integrating Rust with an RTOS such as FreeRTOS or ChibiOS is still a work in progress; especially calling RTOS functions from Rust can be tricky.

We are collecting examples and use cases for this on our issue tracker in issue #62.

A little C with your Rust

Using C or C++ inside of a Rust project consists of two major parts:

- Wrapping the exposed C API for use with Rust
- Building your C or C++ code to be integrated with the Rust code

As C++ does not have a stable ABI for the Rust compiler to target, it is recommended to use the `c` ABI when combining Rust with C or C++.

Defining the interface

Before consuming C or C++ code from Rust, it is necessary to define (in Rust) what data types and function signatures exist in the linked code. In C or C++, you would include a header (`.h` or `.hpp`) file which defines this data. In Rust, it is necessary to either manually translate these definitions to Rust, or use a tool to generate these definitions.

First, we will cover manually translating these definitions from C/C++ to Rust.

Wrapping C functions and Datatypes

Typically, libraries written in C or C++ will provide a header file defining all types and functions used in public interfaces. An example file may look like this:

```
/* File: cool.h */
typedef struct CoolStruct {
    int x;
    int y;
} CoolStruct;

void cool_function(int i, char c, CoolStruct* cs);
```

When translated to Rust, this interface would look as such:

```
/* File: cool_bindings.rs */
#[repr(C)]
pub struct CoolStruct {
    pub x: cty::c_int,
    pub y: cty::c_int,
}

extern "C" {
    pub fn cool_function(
        i: cty::c_int,
        c: cty::c_char,
        cs: *mut CoolStruct
    );
}
```

Let's take a look at this definition one piece at a time, to explain each of the parts.

```
#[repr(C)]
pub struct CoolStruct { ... }
```

By default, Rust does not guarantee order, padding, or the size of data included in a `struct`. In order to guarantee compatibility with C code, we include the `#[repr(C)]` attribute, which instructs the Rust compiler to always use the same rules C does for organizing data within a struct.

```
pub x: cty::c_int,
pub y: cty::c_int,
```

Due to the flexibility of how C or C++ defines an `int` or `char`, it is recommended to use primitive data types defined in `cty`, which will map types from C to types in Rust.

```
extern "C" { pub fn cool_function( ... ); }
```

This statement defines the signature of a function that uses the C ABI, called `cool_function`. By defining the signature without defining the body of the function, the definition of this function will need to be provided elsewhere, or linked into the final library or binary from a static library.

```
    i: cty::c_int,
    c: cty::c_char,
    cs: *mut CoolStruct
```

Similar to our datatype above, we define the datatypes of the function arguments using C-compatible definitions. We also retain the same argument names, for clarity.

We have one new type here, `*mut CoolStruct`. As C does not have a concept of Rust's references, which would look like this: `&mut CoolStruct`, we instead have a raw pointer. As dereferencing this pointer is `unsafe`, and the pointer may in fact be a `null` pointer, care must be taken to ensure the guarantees typical of Rust when interacting with C or C++ code.

Automatically generating the interface

Rather than manually generating these interfaces, which may be tedious and error prone, there is a tool called bindgen which will perform these conversions automatically. For instructions of the usage of bindgen, please refer to the bindgen user's manual, however the typical process consists of the following:

1. Gather all C or C++ headers defining interfaces or datatypes you would like to use with Rust.
2. Write a `bindings.h` file, which `#include "..."`'s each of the files you gathered in step one.
3. Feed this `bindings.h` file, along with any compilation flags used to compile your code into `bindgen`. Tip: use `Builder.ctypes_prefix("cty")` / `--ctypes-prefix=cty` and `Builder.use_core()` / `--use-core` to make the generated code `#![no_std]` compatible.
4. `bindgen` will produce the generated Rust code to the output of the terminal window. This output may be piped to a file in your project, such as `bindings.rs`. You may use this file in your Rust project to interact with C/C++ code compiled and linked as an external library. Tip: don't forget to use the `cty` crate if your types in the generated bindings are prefixed with `cty`.

Building your C/C++ code

As the Rust compiler does not directly know how to compile C or C++ code (or code from any other language, which presents a C interface), it is necessary to compile your non-Rust code ahead of time.

For embedded projects, this most commonly means compiling the C/C++ code to a static archive (such as `cool-library.a`), which can then be combined with your Rust code at the final linking step.

If the library you would like to use is already distributed as a static archive, it is not necessary to rebuild your code. Just convert the provided interface header file as described above, and include the static archive at compile/link time.

If your code exists as a source project, it will be necessary to compile your C/C++ code to a static library, either by triggering your existing build system (such as `make`, `CMake`, etc.), or by porting the necessary compilation steps to use a tool called the `cc` crate. For both of these steps, it is necessary to use a `build.rs` script.

Rust `build.rs` build scripts

A `build.rs` script is a file written in Rust syntax, that is executed on your compilation machine, AFTER dependencies of your project have been built, but BEFORE your project is built.

The full reference may be found here. `build.rs` scripts are useful for generating code (such as via bindgen), calling out to external build systems such as `Make`, or directly compiling C/C++ through use of the `cc` crate.

Triggering external build systems

For projects with complex external projects or build systems, it may be easiest to use `std::process::Command` to "shell out" to your other build systems by traversing relative paths, calling a fixed command (such as `make library`), and then copying the resulting static library to the proper location in the `target` build directory.

While your crate may be targeting a `no_std` embedded platform, your `build.rs` executes only on machines compiling your crate. This means you may use any Rust crates which will run on your compilation host.

Building C/C++ code with the `cc` crate

For projects with limited dependencies or complexity, or for projects where it is difficult to modify the build system to produce a static library (rather than a final binary or executable), it may be easier to instead utilize the `cc` crate, which provides an idiomatic Rust interface to the compiler provided by the host.

In the simplest case of compiling a single C file as a dependency to a static library, an example `build.rs` script using the `cc` crate would look like this:

```
fn main() {
    cc::Build::new()
        .file("src/foo.c")
        .compile("foo");
}
```

The `build.rs` is placed at the root of the package. Then `cargo build` will compile and execute it before the build of the package. A static archive named `libfoo.a` is generated and placed in the `target` directory.

A little Rust with your C

Using Rust code inside a C or C++ project mostly consists of two parts.

- Creating a C-friendly API in Rust
- Embedding your Rust project into an external build system

Apart from `cargo` and `meson`, most build systems don't have native Rust support. So you're most likely best off just using `cargo` for compiling your crate and any dependencies.

Setting up a project

Create a new `cargo` project as usual.

There are flags to tell `cargo` to emit a systems library, instead of its regular rust target. This also allows you to set a different output name for your library, if you want it to differ from the rest of your crate.

```
[lib]
name = "your_crate"
crate-type = ["cdylib"]      # Creates dynamic lib
# crate-type = ["staticlib"] # Creates static lib
```

Building a `C` API

Because C++ has no stable ABI for the Rust compiler to target, we use `C` for any interoperability between different languages. This is no exception when using Rust inside of C and C++ code.

`#[no_mangle]`

The Rust compiler mangles symbol names differently than native code linkers expect. As such, any function that Rust exports to be used outside of Rust needs to be told not to be mangled by the compiler.

`extern "C"`

By default, any function you write in Rust will use the Rust ABI (which is also not stabilized). Instead, when building outwards facing FFI APIs we need to tell the compiler to use the system ABI.

Depending on your platform, you might want to target a specific ABI version, which are documented here.

Putting these parts together, you get a function that looks roughly like this.

```
#[no_mangle]
pub extern "C" fn rust_function() {

}
```

Just as when using `c` code in your Rust project you now need to transform data from and to a form that the rest of the application will understand.

Linking and greater project context.

So then, that's one half of the problem solved. How do you use this now?

This very much depends on your project and/or build system

`cargo` will create a `my_lib.so` / `my_lib.dll` or `my_lib.a` file, depending on your platform and settings. This library can simply be linked by your build system.

However, calling a Rust function from C requires a header file to declare the function signatures.

Every function in your Rust-ffi API needs to have a corresponding header function.

```
#[no_mangle]
pub extern "C" fn rust_function() {}
```

would then become

```
void rust_function();
```

etc.

There is a tool to automate this process, called cbindgen which analyses your Rust code and then generates headers for your C and C++ projects from it.

At this point, using the Rust functions from C is as simple as including the header and calling them!

```
#include "my-rust-project.h"
rust_function();
```

Unsorted topics

Optimizations: the speed size tradeoff

Everyone wants their program to be super fast and super small but it's usually not possible to have both characteristics. This section discusses the different optimization levels that `rustc` provides and how they affect the execution time and binary size of a program.

No optimizations

This is the default. When you call `cargo build` you use the development (AKA `dev`) profile. This profile is optimized for debugging so it enables debug information and does *not* enable any optimizations, i.e. it uses `-C opt-level = 0`.

At least for bare metal development, debuginfo is zero cost in the sense that it won't occupy space in Flash / ROM so we actually recommend that you enable debuginfo in the release profile -- it is disabled by default. That will let you use breakpoints when debugging release builds.

```
[profile.release]
# symbols are nice and they don't increase the size on Flash
debug = true
```

No optimizations is great for debugging because stepping through the code feels like you are executing the program statement by statement, plus you can `print` stack variables and function arguments in GDB. When the code is optimized, trying to print variables results in `$0 = <value optimized out>` being printed.

The biggest downside of the `dev` profile is that the resulting binary will be huge and slow. The size is usually more of a problem because unoptimized binaries can occupy dozens of KiB of Flash, which your target device may not have -- the result: your unoptimized binary doesn't fit in your device!

Can we have smaller, debugger friendly binaries? Yes, there's a trick.

Optimizing dependencies

There's a Cargo feature named `profile-overrides` that lets you override the optimization level of dependencies. You can use that feature to optimize all dependencies for size while keeping the top crate unoptimized and debugger friendly.

Beware that generic code can sometimes be optimized alongside the crate where it is instantiated, rather than the crate where it is defined. If you create an instance of a generic struct in your application and find that it pulls in code with a large footprint, it may be that increasing the optimisation level of the relevant dependencies has no effect.

Here's an example:

```
# Cargo.toml
[package]
name = "app"
# ..

[profile.dev.package."*"] # +
opt-level = "z" # +
```

Without the override:

```
$ cargo size --bin app -- -A
app  :
section               size        addr
.vector_table         1024   0x8000000
.text                 9060   0x8000400
.rodata               1708   0x8002780
.data                    0  0x20000000
.bss                     4  0x20000000
```

With the override:

```
$ cargo size --bin app -- -A
app  :
section               size        addr
.vector_table         1024   0x8000000
.text                 3490   0x8000400
.rodata               1100   0x80011c0
.data                    0  0x20000000
.bss                     4  0x20000000
```

That's a 6 KiB reduction in Flash usage without any loss in the debuggability of the top crate. If you step into a dependency then you'll start seeing those `<value optimized out>` messages again but it's usually the case that you want to debug the top crate and not the dependencies. And if you *do* need to debug a dependency then you can use the `profile-overrides` feature to exclude a particular dependency from being optimized. See example below:

```
# ..

# don't optimize the `cortex-m-rt` crate
[profile.dev.package.cortex-m-rt] # +
opt-level = 0 # +

# but do optimize all the other dependencies
[profile.dev.package."*"]
codegen-units = 1 # better optimizations
opt-level = "z"
```

Now the top crate and `cortex-m-rt` are debugger friendly!

Optimize for speed

As of 2018-09-18 `rustc` supports three "optimize for speed" levels: `opt-level = 1`, `2` and `3`. When you run `cargo build --release` you are using the release profile which defaults to `opt-level = 3`.

Both `opt-level = 2` and `3` optimize for speed at the expense of binary size, but level `3` does more vectorization and inlining than level `2`. In particular, you'll see that at `opt-level` equal to or greater than `2` LLVM will unroll loops. Loop unrolling has a rather high cost in terms of Flash / ROM (e.g. from 26 bytes to 194 for a zero this array loop) but can also halve the execution time given the right conditions (e.g. number of iterations is big enough).

Currently there's no way to disable loop unrolling in `opt-level = 2` and `3` so if you can't afford its cost you should optimize your program for size.

Optimize for size

As of 2018-09-18 `rustc` supports two "optimize for size" levels: `opt-level = "s"` and `"z"`. These names were inherited from clang / LLVM and are not too descriptive but `"z"` is meant to give the idea that it produces smaller binaries than `"s"`.

If you want your release binaries to be optimized for size then change the `profile.release.opt-level` setting in `Cargo.toml` as shown below.

```
[profile.release]
# or "z"
opt-level = "s"
```

These two optimization levels greatly reduce LLVM's inline threshold, a metric used to decide whether to inline a function or not. One of Rust principles are zero cost abstractions; these abstractions tend to use a lot of newtypes and small functions to hold invariants (e.g. functions that borrow an inner value like `deref`, `as_ref`) so a low inline threshold can make LLVM miss optimization opportunities (e.g. eliminate dead branches, inline calls to closures).

When optimizing for size you may want to try increasing the inline threshold to see if that has any effect on the binary size. The recommended way to change the inline threshold is to append the `-C inline-threshold` flag to the other rustflags in `.cargo/config.toml`.

```
# .cargo/config.toml
# this assumes that you are using the cortex-m-quickstart template
[target.'cfg(all(target_arch = "arm", target_os = "none"))']
rustflags = [
  # ..
  "-C", "inline-threshold=123", # +
]
```

What value to use? As of 1.29.0 these are the inline thresholds that the different optimization levels use:

- `opt-level = 3` uses 275
- `opt-level = 2` uses 225
- `opt-level = "s"` uses 75
- `opt-level = "z"` uses 25

You should try `225` and `275` when optimizing for size.

Performing math functionality with #[no_std]

If you want to perform math related functionality like calculating the squareroot or the exponential of a number and you have the full standard library available, your code might look like this:

```
//! Some mathematical functions with standard support available

fn main() {
    let float: f32 = 4.82832;
    let floored_float = float.floor();

    let sqrt_of_four = floored_float.sqrt();

    let sinus_of_four = floored_float.sin();

    let exponential_of_four = floored_float.exp();
    println!("Floored test float {} to {}", float, floored_float);
    println!("The square root of {} is {}", floored_float, sqrt_of_four);
    println!("The sinus of four is {}", sinus_of_four);
    println!(
        "The exponential of four to the base e is {}",
        exponential_of_four
    )
}
```

Without standard library support, these functions are not available. An external crate like `libm` can be used instead. The example code would then look like this:

```rust
#![no_main]
#![no_std]

use panic_halt as _;

use cortex_m_rt::entry;
use cortex_m_semihosting::{debug, hprintln};
use libm::{exp, floorf, sin, sqrtf};

#[entry]
fn main() -> ! {
    let float = 4.82832;
    let floored_float = floorf(float);

    let sqrt_of_four = sqrtf(floored_float);

    let sinus_of_four = sin(floored_float.into());

    let exponential_of_four = exp(floored_float.into());
    hprintln!("Floored test float {} to {}", float, floored_float).unwrap();
    hprintln!("The square root of {} is {}", floored_float,
sqrt_of_four).unwrap();
    hprintln!("The sinus of four is {}", sinus_of_four).unwrap();
    hprintln!(
        "The exponential of four to the base e is {}",
        exponential_of_four
    )
    .unwrap();
    // exit QEMU
    // NOTE do not run this on hardware; it can corrupt OpenOCD state
    // debug::exit(debug::EXIT_SUCCESS);

    loop {}
}
```

If you need to perform more complex operations like DSP signal processing or advanced linear algebra on your MCU, the following crates might help you

- CMSIS DSP library binding
- `constgebra`
- `micromath`
- `microfft`
- `nalgebra`

Appendix A: Glossary

The embedded ecosystem is full of different protocols, hardware components and vendor-specific things that use their own terms and abbreviations. This Glossary attempts to list them with pointers for understanding them better.

BSP

A Board Support Crate provides a high level interface configured for a specific board. It usually depends on a HAL crate. There is a more detailed description on the memory-mapped registers page or for a broader overview see this video.

FPU

Floating-point Unit. A 'math processor' running only operations on floating-point numbers.

HAL

A Hardware Abstraction Layer crate provides a developer friendly interface to a microcontroller's features and peripherals. It is usually implemented on top of a Peripheral Access Crate (PAC). It may also implement traits from the `embedded-hal` crate. There is a more detailed description on the memory-mapped registers page or for a broader overview see this video.

I2C

Sometimes referred to as `I²C` or Inter-IC. It is a protocol meant for hardware communication within a single integrated circuit. See here for more details

PAC

A Peripheral Access Crate provides access to a microcontroller's peripherals. It is one of the lower level crates and is usually generated directly from the provided SVD, often using svd2rust. The Hardware Abstraction Layer would usually depend on this crate. There is a more

detailed description on the memory-mapped registers page or for a broader overview see this video.

SPI

Serial Peripheral Interface. See here for more information.

SVD

System View Description is an XML file format used to describe the programmers view of a microcontroller device. You can read more about it on the ARM CMSIS documentation site.

UART

Universal asynchronous receiver-transmitter. See here for more information.

USART

Universal synchronous and asynchronous receiver-transmitter. See here for more information.